

Geospatial representation models

This deck compares six "representation models" for Earth observation, and then uses them to sharpen the design of our own model for the ocean.

A representation model (also called an embedding model or foundation model) is a neural network that is trained on huge amounts of raw data without labels, and whose job is to turn each place on Earth into a short list of numbers — a vector — that summarises what is going on there. That vector is called an embedding. Once you have good embeddings, many downstream tasks (mapping crops, detecting floods, estimating an ocean current) become much easier because a small model can be trained on top of the embeddings with very few labels.

The word "stencil" runs through the whole deck. It is our term for the piece of the world that feeds one embedding: how large an area around a pixel the model looks at, and how long a stretch of time it summarises. The little drawing on the right shows the idea — a grid of pixels with one output pixel (yellow) and its spatial context (blue), and a timeline with the observations that go into it (orange).

The six models are: AlphaEarth (Google DeepMind), TESSERA (University of Cambridge), OlmoEarth (Allen Institute for AI), TerraMind (IBM and ESA), Prithvi-EO 2.0 (IBM and NASA), and IBM's ocean model, Granite-Geospatial-Ocean. The last third of the deck applies what we learn from them to "Earth 2", our own ocean project.

Sources [Survey notes](#) · [Earth 2 paper \(v8, PDF\)](#)

How to read the stencil diagrams

Every model slide uses the same three-panel picture, so this slide is the legend.

SPACE panel (left). The grid is the tile — the square block of pixels the model processes at once. The blue outline marks the tile edge. The small blue square is one "token": a token is the unit the neural network actually reasons about, usually a small patch of pixels (for example 16 by 16) squashed into one vector. The yellow square is the single output pixel we are asking about. The stacked sheets behind the grid stand for the different data sources — for example optical imagery, radar and elevation — that are laid on top of each other ("co-registered") so that every pixel has a value from each source.

TIME panel (middle). The horizontal axis is time. The shaded band is the "temporal window": the span of time that one embedding summarises. The dots are the individual satellite observations inside that window. If instead there is a single tall bar, the model looks at one moment only — a snapshot.

OUTPUT panel (right). The purple stack represents the embedding vector. "D" is its dimension, the number of entries in the vector — 64, 128, 768 and so on. The caption says whether you get one vector per pixel, per token, or per whole tile.

The four questions on the right are the ones we ask of every model. "Per-pixel or patch" asks whether the model uses neighbouring pixels at all. "Snapshot or period summary" asks whether time is folded into the vector. "Product or model" asks whether the makers hand you pre-computed embeddings, the trained network ("weights"), or both. And "does it touch the ocean" is our own concern: almost all of these models deliberately throw away sea pixels.

Sources [AlphaEarth Foundations paper](#) · [TESSERA paper](#) · [OlmoEarth paper](#) · [TerraMind paper](#) · [Prithvi-EO 2.0 paper](#) · [Granite-Geospatial-Ocean paper](#)

AlphaEarth Foundations (Google DeepMind)

AlphaEarth Foundations is Google DeepMind's model, published in July 2025. Its central idea is the "embedding field": every 10-metre pixel of land on Earth gets a 64-number vector for every year from 2017 to 2024, and Google publishes those vectors as a dataset inside Google Earth Engine.

Inputs: a very wide mix — optical imagery (Sentinel-2, Landsat), radar (Sentinel-1, PALSAR-2), laser altimetry (GEDI), climate reanalysis (ERA5-Land), gravity (GRACE), elevation, land cover and even text descriptions. Over three billion individual observations went in.

Architecture: the encoder is called Space-Time-Precision, STP, and has 15 blocks. The next four slides walk through it. The training uses several losses at once: reconstructing masked-out inputs, keeping the embeddings spread evenly over a sphere ("batch uniformity"), consistency between a teacher and a student network, and matching to text. The deployed network has about 480 million parameters.

The stencil is the key point for us. Output pixel: 10 m. Spatial context: the model uses attention over the whole input frame at coarse resolution, so a pixel's embedding is influenced by everything in its frame — and the supplement tells us how large the frame is: 1.28 km by 1.28 km, 128 pixels a side, for both training and the global inference run (which tiles the world in 960-m squares padded by 160 m on each side). So an AlphaEarth pixel never sees further than about 640 m in any direction. Time: the model takes a "valid period" — a start and end date — and produces a summary of everything observed in it. In principle any window; in practice the published product uses calendar years.

Output: 64 dimensions, normalised to length one (a "unit vector"), stored as 64 bytes. Coverage is land plus coastal and inland water; no open ocean.

Availability is the big caveat: only the dataset is released. The trained network is not, so nobody can run it on new data, new time windows, or the ocean.

"Error reduction ~23.9 %" means: across their 15 evaluation tasks, using AlphaEarth embeddings instead of the next-best method cut the error by about a quarter on average (about 10 % when only ten labels per class are available, about 4 % with one). The per-task numbers are published only as charts, so we do not quote them.

Sources [AlphaEarth Foundations paper](#) · [DeepMind blog](#) · [Satellite Embedding dataset \(Earth Engine\)](#)

AlphaEarth — inside the STP encoder

This slide opens up the encoder. First the symbol: L is the side length of the square input frame, measured in pixels. The main text keeps it symbolic — "given a square input of L pixels a side" — but the supplement fixes it: frames are 128 by 128 pixels, 1.28 km at 10 m, so the three grids below are 8 by 8, 16 by 16 and 64 by 64 cells.

There are three "operators" — three different kinds of neural-network layer — that run in parallel inside every block:

The SPACE operator works on a coarse grid of $L/16$ by $L/16$ cells and uses self-attention, the mechanism from transformers in which every cell can look at every other cell. This gives global context across the frame at low resolution.

The TIME operator works on an $L/8$ grid and uses "time-axial" attention: each location looks only at its own history across all the dated frames. Every frame carries a sinusoidal "timecode" (a numeric encoding of its date), which is how the model knows when things happened.

The PRECISION operator works on the finest grid, $L/2$, with ordinary 3-by-3 convolutions to keep sharp spatial detail.

A "block" is one round of all three operators. After each block, a learned "Laplacian pyramid" rescaling step lets every operator hand its result to every operator of the next block: coarse results are up-sampled, fine results are down-sampled, so that information flows between scales. Repeat for 15 blocks (supplement S2.4, which also gives the widths: 1024 numbers per cell on the space path, 512 on the time path, 128 on the precision path).

Input projectors bring each source's frames down to the $L/2$ grid first. The output head resamples back up to the precision resolution, then to 10 m, and produces the 64-number vector per pixel.

"Teacher/student" refers to the training recipe: one network (teacher) produces targets that a second, identical network (student) learns to match; a third model aligns embeddings with text descriptions in the style of CLIP.

Bottom line: because the space operator uses attention over the whole frame, the effective receptive field is the frame — 1.28 km, so at most about 640 m from the pixel in any direction. What the paper still does not spell out is the exact form of the pyramid resampling and how the three states are merged.

Sources [AlphaEarth Foundations paper](#)

STP, step 1 of 4 — input projectors bring every source to $\frac{1}{2}$ L

Steps 1 to 4 animate the previous slide with the paper's own frame: $L = 128$ pixels at 10 m, so the frame is 1.28 km across (supplement S2.1 and S8.2). Two things in these slides are simplified and the footers say so: we draw 4 blocks where the real network has 15, and the per-year frame counts are typical values rather than the paper's.

Step 1 is about the input projectors. Each data source arrives at its own native resolution over the same 1.28-km footprint: Sentinel-2 at 10 m gives a 128 by 128 grid (drawn coarser here); Landsat at 30 m gives about 43 by 43; ERA5-Land, a weather reanalysis at 9 km, gives just one value for the whole frame. Each source also arrives many times a year — Sentinel-2 roughly every five days, Landsat every 16.

A learned projector per source converts each frame into a "feature map" — a grid of learned numbers rather than raw pixel values — at half the input resolution: 64 by 64 cells of 20 m. This is the only place where resolution is permanently reduced before the output head restores it.

The important detail is that each frame keeps its timestamp: it is carried along as a sinusoidal timecode so the time operator later knows when each observation was made. " $\sum N_i$ frames in, $\sum N_i$ maps out" simply means: however many frames come in from all sources combined, that many feature maps come out — nothing is merged yet.

The yellow pixel is our 10-m output pixel; after this step it lives inside a 20-m feature cell.

Sources [AlphaEarth Foundations paper](#)

STP, step 2 of 4 — three operators, three resolutions

Step 2 shows what one cell can "see" inside each of the three operators, on the same 1.28-km footprint.

SPACE (left): the grid has 8 by 8 cells of 160 m (that is $L/16$ with $L = 128$). The highlighted cell attends to every other cell — the dashed lines. So the space operator gives every cell a view of the whole frame, but only at coarse grain.

TIME (middle): 16 by 16 cells of 80 m. The stacked sheets are the dated frames. The highlighted cell attends only to itself across all the sheets — this is "time-axial" attention. It knows the whole history of one spot, but nothing about the neighbours.

PRECISION (right): 64 by 64 cells of 20 m. The operator is a 3-by-3 convolution, so a cell sees only its eight immediate neighbours — 60 m across — but at the finest grain the block carries. The magnifier shows that 3-by-3 neighbourhood.

The point of running three views at once: none alone is the stencil. Space knows the whole frame coarsely, time knows the whole year at one spot, precision knows the fine local texture. Step 3 is where they are combined.

Sources [AlphaEarth Foundations paper](#)

STP, step 3 of 4 — the pyramid exchange between blocks

This is the sentence the walk-through was built to explain: "after each block, learned Laplacian-pyramid rescaling lets every operator feed every operator of the next block".

On the left are the three outputs of block k — the coarse space map, the medium time map, the fine precision map — each with one feature highlighted. In the middle is the rescaling stage. A Laplacian pyramid is a classic image-processing idea: an image is represented at several resolutions at once, and there are learned operations to move information up (coarse to fine, up-sampling) and down (fine to coarse, down-sampling). Here it is used to translate every map to every resolution.

On the right are the three inputs of block $k+1$. Each now contains all three colours: the blue space feature arrived in the fine map as an 8-by-8 block of 20-m cells, the green precision patch arrived in the coarse map as a fraction of one 160-m cell, the orange time cell landed as 2-by-2 or 4-by-4 blocks. Nine paths in total (three sources times three destinations), then a sum per level — the sum is our reading; the paper says "pass its state to each of the operators" without spelling out the merge.

Repeat 15 times (we draw 4). The consequence for our stencil question: after a few blocks, even the fine 20-m path has a receptive field that spans the whole frame, because it has received the space operator's global view several times — while keeping its own fine grain.

Sources [AlphaEarth Foundations paper](#)

STP, step 4 of 4 — back to 10 m, 64-D per pixel

Step 4 closes the loop. After the last block, the three maps are merged and a final learned spatial resampling takes the result from the L/2 grid (64 by 64 at 20 m) back to L (128 by 128 at 10 m).

The output frame in the middle shows, for one yellow 10-m pixel, what it has inherited: the tinted background stands for the whole frame reaching it through the space path; the dashed squares are the coarse cells it inherited from the space and time operators; the green patch is the fine local texture from the precision path.

Each output pixel becomes a 64-number vector. "Unit vector on S^{63} " is the mathematical way of saying the 64 numbers are scaled so their total length is exactly one — they sit on the surface of a 64-dimensional sphere. The batch-uniformity loss during training spreads the embeddings evenly over that sphere. In the released dataset each vector is stored as 64 bytes (one "int8", a whole number between -128 and 127, per dimension).

"Per valid period": the pair of dates (t_s , t_e) — start and end of the window — was fed into the time operators in every block as timecodes, so the whole frame of vectors is a summary of that window. In the public product the window is one calendar year.

So what is AlphaEarth's stencil? Spatially, the whole 1.28-km frame coarsely (8 by 8 cells of 160 m through the space path) plus a fine local neighbourhood (20-m cells through the precision path); temporally, every dated observation in the window. The number to remember: an AlphaEarth pixel never sees further than about 640 m.

Sources [AlphaEarth Foundations paper](#) · [Satellite Embedding dataset \(Earth Engine\)](#)

TESSERA (University of Cambridge)

TESSERA comes from the University of Cambridge (published June 2025, accepted at CVPR 2026) and is the most radical design in the deck: it uses no spatial context at all.

Inputs: Sentinel-2 optical (10 bands) and Sentinel-1 radar (2 polarisations, VV and VH). Sentinel-1 is radar, so it sees through clouds; Sentinel-2 is optical.

Architecture: for each 10-m pixel separately, a small transformer reads the pixel's own time series — 40 dates sampled from one calendar year for each sensor — and a GRU (a type of recurrent network) pools them into one vector. The two sensors have their own branches which are fused at the end. The encoder is only about 46 million parameters; a much larger "projector" is used only during training and then discarded. Training uses Barlow Twins, a self-supervised objective that asks two differently-sampled views of the same pixel-year to produce the same embedding while keeping the embedding's dimensions non-redundant — no reconstruction of inputs is involved.

Stencil: strictly one 10-m pixel; neighbours never enter. One calendar year; dates are irregular, and the model is told the day of year of each observation. If a pixel has fewer than 40 clear observations, dates are sampled with replacement to fill the 40 slots.

Output: 128 numbers per pixel per year, stored as int8. Years 2017–2025, land only (water masked). Everything is released: weights (CC0, the most permissive licence), code (MIT), and precomputed embeddings through the GeoTessera service.

Results quoted are with the embeddings frozen and only a small head trained on top: PASTIS-R crop segmentation mIoU 50.7, TreeSatAI tree-species F1 78.0, canopy height error 12.2 m. mIoU is "mean intersection over union", the standard segmentation score (100 = perfect overlap). F1 is the balanced average of precision and recall for classification.

Why it matters for us: TESSERA proves that a per-pixel temporal encoder with zero spatial context is competitive on land. That is the opposite extreme from our 3-by-3 patch, and a useful reference point.

Sources [TESSERA paper](#) · [TESSERA code](#) · [GeoTessera embeddings](#)

OlmoEarth (Ai2)

OlmoEarth is the Allen Institute for AI's family of open models, first released November 2025; version 1.1 (May 2026) merged Sentinel-2's band groups into a single token to cut compute, and version 1.2 added rotary position encodings. It is a "space-time ViT": a Vision Transformer that tokenises not just space but also time and sensor band groups.

Inputs: Sentinel-1, Sentinel-2 and Landsat-8 monthly time series, all resampled to 10 m, plus static map layers (OpenStreetMap, land cover, crop data, elevation, canopy height) used as extra inputs and as prediction targets.

Architecture: the tile is 2.56 km square (256 by 256 pixels at 10 m). Tokens are flexible patches of 1 to 8 pixels — so 10 to 80 m — and up to 12 monthly steps are tokenised separately. Attention runs across space, time and modality together. The training objective, "Latent MIM Lite", is masked modelling: hide some tokens, predict them in the embedding space rather than as pixels, plus contrastive terms. Four sizes: Nano (1.4 M parameters), Tiny (6 M), Base (89 M), Large (308 M).

Output: one vector per patch per month (768 dimensions for Base, 1024 for Large) plus a pooled vector for the tile. There is no global precomputed product; instead the "OlmoEarth Studio" computes embeddings on demand for any region you draw.

Results: in the authors' own benchmark the model is best on 15 of 24 tasks with frozen embeddings and 19 of 29 when fine-tuned; on five tasks a fine-tuned OlmoEarth matches or beats AlphaEarth.

Licence: open weights, but under a custom licence that forbids some uses (military, extractive industries).

For us: OlmoEarth is the closest analogue to a general space-time transformer we could actually retrain — open weights, monthly tokens, a masked latent objective — but its pretraining sample is land-biased.

Sources [OlmoEarth paper](#) · [Ai2 OlmoEarth blog](#) · [OlmoEarth pretraining code](#) · [OlmoEarth v1.1](#) · [OlmoEarth v1.2](#)

TerraMind (IBM Research + ESA Φ-lab)

TerraMind is a joint IBM Research and ESA Φ-lab model (April 2025, ICCV 2025). Its distinguishing feature is that it is generative and "any-to-any": given some modalities it can produce the missing ones.

Inputs come in two kinds. Pixel-level: Sentinel-2 optical, Sentinel-1 radar, elevation, RGB, vegetation index (NDVI). Token-level: land-use class maps, text captions, geographic coordinates. All are aligned on 10-m, 224-pixel tiles.

Architecture: each modality has its own "tokenizer" — a small network that turns its raster into discrete codes from a 16,000-entry codebook (an FSQ-VAE, a kind of quantising autoencoder). The main encoder-decoder then learns to predict masked codes of one modality from the others. "Thinking in Modalities" (TiM) is the trick of generating a missing modality — say land cover from optical — and feeding it back in as extra input. Trained on 500 billion tokens from about 9 million TerraMesh samples.

Stencil: tile 2.24 km (224 pixels at 10 m); token 16 pixels = 160 m, so 14 by 14 tokens per modality. One co-registered timestamp per sample — every modality comes from the same date and place; there is no multi-temporal fusion in pretraining.

Output: 196 tokens per modality, 768 dimensions for the Base model and 1024 for Large, or one merged vector per tile, plus generated rasters. No precomputed embedding archive; weights are Apache 2.0 (fully open) on Hugging Face.

Results: on the PANGAEA benchmark (a standard suite of segmentation tasks) TerraMind Large scores 59.6 mean IoU, the only foundation model to beat U-Nets trained from scratch on that suite.

Ocean: the training data is stratified by land ecoregion; there are no marine modalities and open-ocean behaviour is untested.

Sources [TerraMind paper](#) · [TerraMesh dataset paper](#) · [TerraMind code](#) · [TerraMind weights \(Hugging Face\)](#)

Prithvi-EO 2.0 (IBM + NASA)

Prithvi-EO 2.0 is IBM and NASA's second-generation Earth observation model (December 2024). It matters here for two reasons: it is a widely used open baseline, and IBM's ocean model on the next slide reuses its recipe.

Inputs: HLS, the Harmonized Landsat-Sentinel-2 product, at 30 m with six spectral bands (blue, green, red, near-infrared, and two shortwave-infrared bands). 4.2 million training samples from 2014–2023. Importantly for us, sea-only tiles were explicitly removed using a cloud/water mask, and Greenland was excluded.

Architecture: a masked autoencoder (MAE) — the network sees a tile with 75 % of its patches hidden and learns to reconstruct them. Two sizes: 300 million parameters (a "ViT-Large", 1024-dimensional embeddings, 24 layers) and 600 million (ViT-Huge, 1280 dimensions, 32 layers). The "TL" variants additionally receive the acquisition date and latitude/longitude as extra encodings.

Stencil: tile 224 pixels at 30 m = 6.72 km. Token 16 pixels = 480 m for the 300 M model, 14 pixels = 420 m for the 600 M model (we checked the published configuration files). Time: four frames, each 1–6 months apart to capture seasons; the "tubelet depth" is 1, meaning each frame gets its own grid of tokens rather than frames being merged into one token — time is fused only inside attention. Single-frame use is also supported.

Output: one vector per token per frame; no precomputed product. The typical way to use it is full fine-tuning — retraining the encoder together with a small decoder for your task — through IBM's TerraTorch toolkit.

Results: on GEO-Bench (a 12-task standard benchmark) the 600 M-TL model is the best on average, about 8 % above version 1.0 and ahead of six other foundation models; the per-dataset scores are published as charts, so we quote no single number. In May 2026 the model was deployed on a satellite for onboard flood and cloud detection.

Sources [Prithvi-EO 2.0 paper](#) · [Prithvi-EO 2.0 code](#) · [Prithvi weights and configs \(Hugging Face\)](#)

Granite-Geospatial-Ocean (IBM • STFC • PML • Exeter)

This is IBM's Prithvi-based ocean model, developed with the UK's STFC Hartree Centre, Plymouth Marine Laboratory and the University of Exeter, released October 2025. The code lives in the `ibm-granite` GitHub organisation; we read its configuration files directly to get the numbers below.

Inputs: Sentinel-3 OLCI, the ocean-colour instrument, 16 spectral bands at 300 m, plus sea-surface temperature from the SLSTR instrument — 17 channels in total. About 512,000 tiles from 2017–2021 (the paper also quotes 470,000 for training plus 50,000 for validation), sampled evenly across 83 "Longhurst provinces" (a standard division of the ocean into biogeochemical regions) and across months.

Architecture: the Prithvi masked-autoencoder recipe with the network shrunk to fit the coarser sensor: embedding size 512, 12 layers, about 50 million parameters, trained to reconstruct masked patches under a root-mean-square-error loss. The authors report that larger networks gave no gain.

Stencil: tile 42 by 42 pixels at 300 m = 12.6 km square. Token 2 by 2 pixels = 600 m, so 21 by 21 = 441 tokens. `num_frames = 1`: a single acquisition, no temporal stacking — the four-frame option of Prithvi is switched off. For fine-tuning, each ship measurement is paired with cloud-free imagery from a six-day window centred on it.

Output: 441 vectors of 512 dimensions per tile. Downstream, a small regression head is trained to predict chlorophyll-a (a proxy for phytoplankton) and primary production (how much carbon the phytoplankton fix).

Results: the numbers are RMSE — root-mean-square error — in log10 units, because these quantities vary over orders of magnitude. With the foundation model: 0.14 for chlorophyll, versus 0.16 training the same network from scratch; 0.39 versus 0.42 for primary production. The gains are small and within the fold-to-fold spread; the authors' stronger claim is that the advantage grows when labels are scarce.

Caveats: surface-only, optical (so cloud-limited), one snapshot in time. It is a model of ocean colour — biology at the surface — not of the ocean's physical state.

Sources [Granite-Geospatial-Ocean paper](#) • [ibm-granite/geospatial \(code + config\)](#) • [Granite-Geospatial-Ocean weights](#)

Stencils side by side

This chart puts all six models — and our own — on two axes that summarise the whole deck.

Horizontal axis (logarithmic): the spatial extent of the input that shapes one embedding, from 10 m to 1000 km. Vertical axis (categorical): how much time one embedding summarises, from a single snapshot at the bottom to a full year of all sources at the top.

Reading the points: TESSERA sits at 10 m on the far left — a per-pixel output with no spatial context at all — and AlphaEarth at 1.28 km, the frame each of its 10-m pixels is computed from; both sit at the top because both fold a whole year into one vector. TerraMind and Granite are single snapshots at 2–13 km tile scale. Prithvi stacks four dates; OlmoEarth stacks twelve monthly steps.

Earth 2, our codec, is the purple dot at about 84 km: each channel token sees a 3-by-3 neighbourhood of quarter-degree cells (a quarter degree is about 28 km) at one monthly or five-day mean. It is a snapshot stencil roughly seven times wider than the widest Earth-observation tile here (Granite, 12.6 km). The hollow dot is the proposed "pixel-year" experiment from a later slide, which would move us up to the "one year folded" row.

Two design axes separate all these models: per-pixel versus patch, and snapshot versus period summary. Every later recommendation in the deck comes back to these two axes.

Sources [AlphaEarth Foundations paper](#) · [TESSERA paper](#) · [OlmoEarth paper](#) · [TerraMind paper](#) · [Prithvi-EO 2.0 paper](#) · [Granite-Geospatial-Ocean paper](#)

Summary I — what goes into each embedding

A reference table of what goes into each embedding, condensing the model slides. Column by column:

Inputs: the sensors and data sources. S-1 and S-2 are Sentinel-1 (radar) and Sentinel-2 (optical); HLS is the harmonised Landsat–Sentinel product; OLCI and SLSTR are the Sentinel-3 ocean-colour and temperature instruments.

Spatial stencil: what area feeds one embedding — the output pixel size, the token size (the patch the network reasons about) and the tile size (the block it processes at once).

Temporal stencil: whether the model summarises a period (AlphaEarth, TESSERA), stacks dated frames (OlmoEarth, Prithvi), or takes a single snapshot (TerraMind, Granite).

Output: the vector dimension and what one vector describes — a pixel-year, a token, a tile.

The table is designed to be read across a row to reconstruct a model's stencil, or down a column to compare one design choice.

Sources [AlphaEarth Foundations paper](#) · [TESSERA paper](#) · [OlmoEarth paper](#) · [TerraMind paper](#) · [Prithvi-EO 2.0 paper](#) · [Granite-Geospatial-Ocean paper](#)

Summary II — what each project hands you

The companion table: what each project actually hands you, which is often as important as the architecture.

Open weights: whether the trained network is downloadable, so you can run it on your own data or fine-tune it. AlphaEarth is the exception — no weights, no code.

Precomputed embeddings: whether a ready-made global product exists. AlphaEarth and TESSERA yes; OlmoEarth on demand through its Studio; the others no.

Licence: CC-BY and CC0 are Creative Commons licences (attribution required, or no restrictions); Apache 2.0 and MIT are permissive software licences; OlmoEarth's custom licence has use restrictions. Prithvi's licence is stated differently on Hugging Face and GitHub, hence "verify".

Ocean: green means open ocean is in the training data, red means it is masked out or not targeted. Only Granite is green among the six.

Intended use: "frozen embeddings" means you keep the network fixed and train only a small head on top; "fine-tuning" means you retrain the network for your task; kNN is k-nearest-neighbours, the simplest possible classifier, used to test how good raw embeddings are.

Sources [Satellite Embedding dataset \(Earth Engine\)](#) · [GeoTessera embeddings](#) · [OlmoEarth pretraining code](#) · [TerraMind weights \(Hugging Face\)](#) · [Prithvi weights and configs \(Hugging Face\)](#) · [Granite-Geospatial-Ocean weights](#)

Who embeds the ocean — and how deep?

This slide answers a question we kept returning to: which representation models have ever seen the sea in their training data, and how deep do they go?

Earth 2 (ours) is in the first row because it is the only entry whose inputs include the ocean interior — temperature and salinity at depth from Argo floats (the global fleet of autonomous profiling instruments), plus sea-surface height, currents and wind at the surface. Our stencil is a 3-by-3 patch of quarter-degree cells, one monthly or five-day snapshot, with 24 months of history in the second-stage forecasting model. Output: 64 numbers per pixel-month.

Granite-Geospatial-Ocean (previous slide) and OceanSAR are the two published encoders trained deliberately on sea pixels — but both are surface-only. Granite sees ocean colour and temperature; OceanSAR (from the company Galeio) sees Sentinel-1 radar "wave mode" images of the sea surface, about 12 million of them, and predicts sea state, wave height and wind speed.

Copernicus-FM (TU Munich) covers "the whole land surface and near-land ocean" on a quarter-degree grid — coastal seas are incidental. Weather and climate models (Prithvi WxC, Aurora, ClimaX) contain ocean grid cells only because reanalysis fields are global; the sea is background, not the learning target. Location encoders such as SatCLIP work on coordinates, not ocean state.

Verdict: nothing else embeds the ocean interior through time. Forecast systems such as GLONET or Samudra learn ocean dynamics but publish forecasts, not reusable embeddings.

Sources [Granite-Geospatial-Ocean paper](#) · [OceanSAR-1 paper](#) · [OceanSAR-2 paper](#) · [Copernicus-FM paper](#) · [Aurora \(Nature 2025\)](#) · [AIFS surface ocean \(ECMWF 2026\)](#) · [Earth 2 paper \(v8, PDF\)](#)

Four interacting spheres — who embeds which

Chris's framing of the long-term goal: the Earth system has four interacting "spheres", and eventually we want one representation that spans them all. This slide maps who covers which today and proposes an order for joining them.

a) Atmospheric weather: covered by forecast models (Prithvi WxC, Aurora, GraphCast and others) that produce states, not reusable embeddings. Native timescale hours to days; the atmosphere is predictable for about ten days.

b) Ocean weather: the surface is covered by OceanSAR, Granite and ECMWF's 2026 AIFS ocean extension; the interior only by Earth 2. Timescales run from days at the surface to decades in the deep. A physical fact drives everything: the top two and a half metres of ocean hold as much heat as the entire atmosphere above it, so the ocean is the slow, remembering partner.

c) Ocean biosphere: phytoplankton, ocean colour, carbon uptake. Granite plus task-specific carbon-flux models (SOM-FFN, CSIR-ML6, CANYON-B). Timescales: blooms in weeks, seasons, decadal trends.

d) Land biosphere: the entire mainstream of Earth observation models, all of which fold a year and none of which touch the sea.

Nobody represents two spheres jointly. The proposed sequence: start from the ocean interior (longest memory, least covered by anyone else — what we already have), add ocean colour (same grid, Granite's recipe exists, and it closes the physics-to-biology loop through mixed-layer depth, light and temperature), then treat the atmosphere as a forcing (already in our channels as wind and fluxes), and add land last.

Three design consequences follow: each sphere needs its own speed-and-memory "cone" (introduced on slide 26); biospheres run on the calendar so they want year-summary tokens while physics wants dated snapshots; and the observing systems differ in shape (grids, sparse profiles, cloud-gapped swaths, 10-m tiles), which our distinction between "hidden from the model" and "never measured" tokens is built to handle.

Sources [AIFS surface ocean \(ECMWF 2026\)](#) · [Samudra](#) · [SamudrACE](#) · [Granite-Geospatial-Ocean paper](#) · [ESFM](#) · [Earth 2 paper \(v8, PDF\)](#)

Head-to-head I — frozen embeddings, one protocol

The first of four slides with actual benchmark numbers, transcribed from the papers. This one uses the widest comparison available: Table 2 of the OlmoEarth paper, which evaluates many models with a single frozen-embedding protocol.

"Frozen" means the encoder is not retrained; only a trivial classifier sits on top. kNN (k-nearest neighbours) assigns a test image the majority label of its 20 nearest training embeddings — the purest test of embedding quality. "Linear probe" fits a single linear layer, used for segmentation and multi-temporal tasks.

Columns are benchmark datasets: BigEarthNet and So2Sat (land-cover classification), EuroSAT (scene classification), CropHarvest-Togo (crop vs non-crop), m-cashew and SA-crop (crop segmentation), PASTIS (crop parcels through time), MADOS (marine debris and oil), Sen1Floods11 (flood mapping from radar). Scores are accuracy or micro-F1 for classification and mIoU for segmentation — all 0 to 100, higher is better.

Shaded rows are the models in this deck; grey rows are other models for context. Green is the best in each column across all rows.

What it says: on frozen features OlmoEarth leads the classification columns by large margins (So2Sat 68 vs 47 for TerraMind vs 35 for Prithvi), TerraMind leads BigEarthNet and the cashew and SA-crop segmentation, and Prithvi trails on every classification column. TESSERA appears once, on CropHarvest-Togo, within one point of OlmoEarth. AlphaEarth is absent from this table because its embeddings are annual and could not be run under this protocol.

Caveat, which applies to every table in this section: it was authored by one of the contestants.

Sources [OlmoEarth paper](#) · [TESSERA paper](#)

Head-to-head II — fine-tuning and PANGAEA

Two more tables. The top one is OlmoEarth's Table 3: the same models, now fully fine-tuned — the whole network is retrained for each task (the encoder is unfrozen after the first 20 % of training). The bottom one is TerraMind's Table 6 on the PANGAEA benchmark, a community suite of about eleven segmentation, change-detection and regression tasks, evaluated with a frozen encoder plus a standard segmentation decoder (UPerNet).

Top table: once fine-tuned, the gap between models closes. TerraMind Large and OlmoEarth Large split the columns roughly evenly, and Prithvi-EO 2.0 (600 M) sits one to three points behind on the classification columns but eight to thirteen points behind on the PASTIS and MADOS segmentation columns. The last row is the same OlmoEarth architecture trained from random initialisation — no pretraining at all. The distance between that row and the others (10 to 37 points) is the value of pretraining itself.

Bottom table: mIoU per dataset plus an average and an average rank. TerraMind Large is the only foundation model whose average (59.6) beats a U-Net trained from scratch (57.6). Prithvi appears only as version 1.0 and comes last. HLS Burn Scars, FiveBillionPixels, DynamicEarthNet, SpaceNet 7 and AI4SmallFarms are the other PANGAEA datasets.

Not shown: Prithvi-EO 2.0's own GEO-Bench table, because none of the other models in this deck appear in it.

Sources [OlmoEarth paper](#) · [TerraMind paper](#) · [PANGAEA benchmark](#)

Head-to-head III — the two embedding products

This slide compares the two "products" — the models whose main deliverable is a precomputed global embedding layer: AlphaEarth and TESSERA.

Left table, from the TESSERA paper: frozen embeddings plus a light head, on six tasks. F1 and mIoU are higher-is-better (arrows up); RMSE — root-mean-square error, in tonnes per hectare for biomass and metres for canopy height — is lower-is-better (arrows down). TESSERA beats AlphaEarth on five of six, with the largest margins on pixel-wise crop tasks where a per-pixel time series matters most (Austrian crop F1 82 vs 56). AlphaEarth edges PASTIS-R by 0.4 points. Both products beat every frozen ViT-style model by wide margins.

Top-right table, from the OlmoEarth paper: AlphaEarth versus OlmoEarth under three protocols. With both frozen, results are mixed. Fine-tuned OlmoEarth wins all five tasks — and the "n/a" row is the point: AlphaEarth cannot be fine-tuned, because only the embeddings, not the network, are released. LFMC is live fuel moisture content, an L1 (absolute) error, lower is better.

Bottom-right, an independent study: mapping Local Climate Zones in Bern with an Attention U-Net fed either raw Sentinel composites, AlphaEarth, or TESSERA. The two products are within one point of each other and two to five points above raw imagery.

AlphaEarth's own paper reports a 23.9 % average error reduction versus the next-best method across 15 evaluations, losing only on land-use change; its per-task values are published as charts without numbers, which is why they are not tabulated here.

Sources [TESSERA paper](#) · [AlphaEarth vs TESSERA for Local Climate Zones](#) · [OlmoEarth paper](#) · [AlphaEarth Foundations paper](#)

Head-to-head IV — the ocean encoders

The two ocean encoders cannot be compared with the land models — no shared benchmark exists — so this slide shows each against its own baselines.

Left: Granite-Geospatial-Ocean, Table 1. Five-fold cross-validated RMSE in log10 units for chlorophyll-a and primary production, comparing the fine-tuned foundation model with the same network trained from scratch and with a random forest on raw pixel values. The foundation model is best in both columns, but the margins (0.02 and 0.03) are smaller than the spread between folds. The authors' stronger claims are label efficiency (the advantage grows as training labels shrink) and better spatial pattern (SSIM — structural similarity — of 0.88 versus 0.82 against the operational product).

Right: OceanSAR-1, kNN on frozen features against other SAR foundation models. TenGeoP is a ten-class geophysical-phenomena dataset (accuracy, higher is better); wave height and wind speed are RMSE in metres and metres per second (lower is better). Ocean-only pretraining beats land-trained SAR models by 9 to 23 points on classification and cuts wind-speed error by 30 to 45 %.

Bottom box: the only thing the ocean and land models share is the protocol — frozen features plus a small head, and "foundation model versus same network from scratch" as the control for the value of pretraining. That is also exactly what our attribution matrix does: our pixel and patch codecs versus raw data at matched receptive field. Our pretraining margin (0.617 vs 0.613 correlation for a single pixel; 0.672 vs 0.659 for the 3-by-3 patch) is the same kind of statement as Granite's 0.14 versus 0.16 — a small one.

Sources [Granite-Geospatial-Ocean paper](#) · [OceanSAR-1 paper](#) · [OceanSAR-2 paper](#)

Who wins where — a digest of the tables

A one-page digest of the four head-to-head slides: for each evaluation protocol and metric family, how the models in this deck rank, with the margin and the source.

Reading down: with frozen features OlmoEarth leads classification and shares segmentation with TerraMind; fine-tuned, TerraMind Large and OlmoEarth Large tie with Prithvi behind; on PANGAEA TerraMind is the only model above a from-scratch U-Net; GEO-Bench-2 (an independent 2025 benchmark that publishes ranks, not scores) places Prithvi 600M-TL fifth and TerraMind Large sixth of fourteen; among the two products TESSERA beats AlphaEarth on five of six tasks in TESSERA's own table but they tie in an independent study; a fine-tuned OlmoEarth beats AlphaEarth everywhere because AlphaEarth cannot be fine-tuned; and the ocean models are measured only against themselves.

The two structural caveats at the bottom are the honest summary. First, no benchmark contains all six models — the widest single table has five and omits AlphaEarth, and the only independent multi-model source has two. Second, no public benchmark has an ocean-state, transport or interior task, so none of these rankings transfer to our problem. What transfers is the protocol: frozen probe versus fine-tune, and a from-scratch control at matched receptive field, which our attribution matrix already implements.

Sources [OlmoEarth paper](#) · [TerraMind paper](#) · [Prithvi-EO 2.0 paper](#) · [TESSERA paper](#) · [GEO-Bench](#) · [GEO-Bench-2](#) · [AlphaEarth vs TESSERA for Local Climate Zones](#) · [Granite-Geospatial-Ocean paper](#) · [OceanSAR-1 paper](#)

What this means for our ocean representation work

Four takeaways for Earth 2, each tied to something measured.

Axis 1, pixel versus patch. TESSERA shows a per-pixel encoder with no spatial context is competitive on land; every other model buys context with a patch grid. We have already priced this axis in our attribution matrix: reading the RAPID transport with a single-pixel codec gives a correlation of 0.617 (raw data 0.613); with the 3-by-3 patch, 0.672 averaged over two seeds (raw 0.659). Taken at face value that would make the neighbourhood worth about +0.05 and pretraining about +0.01. (RAPID is the mooring array at 26.5°N that measures the Atlantic overturning circulation; "correlation" is how well our read-out tracks it.) Two cautions. These are probe numbers, whose seed-to-seed spread in our own record runs from 0.04 to 0.25, so the +0.05 sits at the edge of the noise band and the +0.01 inside it — the honest reading is "the neighbourhood probably matters; pretraining is unresolved". And on 2 September the paper was reset, because its rolled forecast numbers came from second-stage heads that had seen the held-out years; these probe numbers are unaffected by that (the reset paper keeps them in its appendix), but no forecast number from before 2 September may be quoted anywhere.

Axis 2, snapshot versus period. TerraMind and Granite are snapshots; Prithvi and OlmoEarth stack dated frames; AlphaEarth and TESSERA fold a full year into one vector. Our first-stage codec is a snapshot of a monthly (or five-day) mean, and the month-block codec folds only about six five-day bins. The year-folded design is the axis we have not tested — hence the proposed experiment on the next slide.

Product versus model. AlphaEarth is a dataset you cannot re-run; TESSERA and OlmoEarth ship both. For the ocean nothing precomputed exists at all.

The gap. Only two published encoders are trained on sea pixels — Granite and OceanSAR — and both are surface, single-instant models. To our knowledge no published model embeds the gridded ocean state through time. That is the territory our codec occupies.

Sources [Earth 2 paper \(v8, PDF\)](#) · [ml/Claude.md \(replication rules, §3b\)](#) · [TESSERA paper](#) · [AlphaEarth Foundations paper](#)

Proposed arm: a TESSERA-style pixel-year codec

A concrete experiment proposal, framed honestly. An earlier draft of this deck suggested a "1-by-1 pixel-only control"; it turned out our paper already contains it (the attribution matrix has both single-pixel and 3-by-3 versions of raw and codec inputs). So the spatial question has been measured, at the resolution probe numbers allow; what TESSERA adds that we have not tried is the temporal fold.

What TESSERA does that we don't: it produces one embedding per pixel-YEAR from about 40 dated observations of one pixel, using a temporal transformer and day-of-year encodings, trained with Barlow Twins (no reconstruction) and quantised to int8 during training. Our first-stage vector z describes one pixel-MONTH; a second-stage model then runs over the sequence of monthly vectors.

Arm A — the pixel-year codec: an encoder that reads the 12 (or 24) pixel-months of one quarter-degree cell at once, with month-of-year tokens, and emits one 64-number vector per pixel-year. We keep the patch size at 1 so that only the time axis changes. Token count about 12 times (channels + 2), roughly 310 at 24 channels — the same order as our existing 282-token month-block codec, so it fits the current training recipe.

Arm B — swap the training objective: same encoder, but Barlow Twins on two random subsets of dates from the same pixel-year, instead of masked reconstruction.

Controls: the average of the 12 monthly vectors (does folding beat pooling?), the current stage-2 pathway, and the raw 12-month stack fed to the same attention head (the end-to-end control from the attribution matrix). At least two codec seeds, because head numbers from one seed have proven noisy.

Read-outs: our probe ladder to the RAPID transport (correlation and RMSE in Sverdrups), the fraction of variance lost on the round trip, and rolled skill under the corrected protocol — the mean skill against climatology and against damped persistence at each lead, with trained and held-out longitudes reported separately; the old "corridor AUC" name was retired on 2 September 2026 when the paper was reset. Success means the folded vector beats the control beyond block-bootstrap intervals, with at least three seeds.

Sources [TESSERA paper](#) • [Earth 2 paper \(v8, PDF\)](#) • [ml/CRAUDE.md \(replication rules, §3b\)](#) • [E-069 plan — two stencils, one cone](#)

A velocity-specific dependency cone per channel

This slide introduces the physical idea Chris proposed: what can influence a pixel Δt ahead is bounded by how fast each process propagates and how long it remembers.

Notation first. t is "now", the last observed step. Δt is the forecast step — we predict the target at $t + \Delta t$. ℓ (a script L) is the lag of an input: it was observed at $t - \ell$, so it sits $\Delta t + \ell$ before the target; $\ell = 0$ is the newest input. v is a process's signal speed; τ (tau) is its memory — how long it stays predictable.

The model: for a process p , an input at lag ℓ can matter only if it is close enough to have travelled to the target in time, $|\Delta x| \leq v \cdot (\Delta t + \ell)$, and only if the process still remembers, $\Delta t + \ell \leq \tau$. The first condition is the domain-of-dependence or "light-cone" argument from numerical physics (the CFL condition); the second says that beyond a process's predictability its history is noise.

The chart is log-log: time separation on the horizontal axis (1 day to 10 years), distance on the vertical (10 km to 20,000 km), so a constant speed is a straight diagonal. Each process is a region: the atmosphere (amber) is wide but shallow — 10 m/s but only 10 days of memory; the surface ocean (blue) is narrow but months deep — 0.1 m/s, 8 months; the interior (teal) is narrower still but years deep — 0.03 m/s. The dashed line is the Kelvin-wave speed along coasts and the equator, an exception that is fast but anisotropic.

The union rule: a channel forced by several processes has the union of their cones. Sea-surface temperature is forced by the atmosphere (heat flux) and by currents (advection), so its region is wide-and-shallow plus narrow-and-deep — an L-shape, not a cone.

Where our stencils sit: a fixed stencil is a cylinder — same reach at every lag. Our ring-8 at 222 km per month is an ocean-advection speed (0.085 m/s); our sunflower-89 at 4444 km is 1.7 m/s, inside the atmosphere's cone at a one-month step. The dashed vertical line marks the 24 months of history our second-stage model sees.

Sources [Dependency-Cone Explorer](#) • [Chelton et al. 2011, global eddies](#) • [Courant, Friedrichs & Lewy 1928](#)

What the cone predicts — and how to test it on our stencil

The cone model applied channel group by channel group, with the useful reach per monthly step, the useful history, and a design consequence for each.

Wind, pressure and fluxes: the atmosphere moves at 10 m/s and forgets in 5–10 days. Its causal reach in one month is the whole globe, but because its memory is shorter than our step, monthly wind is not predictable from its own history — only through slow ocean modes that force it. So treat it as a wide, shallow forcing: large stencil, no depth. At the five-day step the memory and the step are similar, so wind becomes partly self-predictable — a gain only a pentad model could show, and one that has not been shown: the earlier pentad rolls were withdrawn with the paper reset.

Sea-surface and mixed-layer temperature: advected at 0.1–0.3 m/s (260–800 km per month), forced by the atmosphere, remembered for 3–12 months (including "re-emergence" — anomalies that sink in autumn and reappear next spring). It needs both the wide wind stencil and months of history: the one group that justifies the full cylinder.

Interior temperature and salinity: 0.02–0.1 m/s, memory of years to decades. Per monthly step the useful reach is 50–300 km, which a 3-by-3 patch at quarter degree already covers; our 4444-km stencil is 15–90 times too wide for one step. Small reach, long history — the concrete answer to the open question "can reach come down?" from the project's dependency-cone note.

Sea-surface height: slow Rossby waves in the interior (0.03 m/s at mid-latitudes), fast Kelvin and coastal waves along boundaries (1–3 m/s). An anisotropic stencil — an arm along coasts, narrow elsewhere.

The transport target (RAPID): the overturning transport is set by the east-west density difference across the basin ("thermal wind"), which means its cone is the whole section — and why a read-out must see the whole section rather than a spatial mean, which would average away exactly that east-west contrast (an argument from the mechanism; the measured pooled-versus-unpooled gap is a two-interval comparison our own rules do not allow as evidence).

Bottom left: four falsifiable predictions, each an ablation to run under the corrected roll protocol of the reset paper (none has been run; the earlier rolled numbers were withdrawn). Bottom right: where the simple picture needs care — v is the fastest signal speed, not the mean flow; τ is predictability, not causality; cones are tilted, not symmetric; monthly averaging turns anything faster than the step into forcing.

Sources [Dependency-Cone Explorer](#) · [Chelton et al. 2011, global eddies](#) · [Earth 2 paper \(v8, PDF\)](#) · [ml/CLAUDE.md \(replication rules, §3b\)](#)

Worked example — the surface current at one pixel

A concrete walk-through of the cone idea for one target: the surface current (u , v) at a quarter-degree pixel, one month ahead. Each row is a driver — something that pushes the current around — with its mechanism and its two numbers, speed and memory, plus a correlation length.

Wind stress acts through Ekman transport (the wind drags the surface layer, which turns at right angles because of Earth's rotation), within about a day. Because its 10-day memory is shorter than the monthly step, only the newest input (lag 0) is useful, and only as forcing that is not itself predictable.

Air temperature acts through heat flux, which changes mixed-layer temperature, which changes density gradients, which drive geostrophic currents. This is the subtle row: the atmosphere forgets in 10 days, but the mixed layer remembers the heat it received for months and carries it along at ocean speed — so the channel inherits the ocean's memory (6 months) and speed (0.15 m/s). Its reach is the 1,500-km atmospheric correlation length plus what the ocean advects.

Ocean surface temperature drives currents through horizontal density gradients ("thermal wind") — which is why neighbours are mandatory: a gradient does not exist inside one pixel.

Interior temperature and salinity: same mechanism, integrated from depth; at 0.03 m/s (78 km per month) the 3-by-3 patch already covers lag 0.

Sea-surface height sets the pressure gradient; slow Rossby waves in the interior plus a fast Kelvin arm along coasts.

Own history: momentum persistence and eddies. Individual eddies live four months and longer and drift westward at a few centimetres per second, but at a fixed pixel the current decorrelates in weeks to a few months because they pass through — that decorrelation time, about three months, is the memory that matters here.

The "two rules" box states the arithmetic used throughout: reach at lag ℓ is the larger of $v \cdot (\Delta t + \ell)$ and the correlation length, capped at 10,000 km; a lag is useful if $\Delta t + \ell \leq \tau$. There is one deliberate exception, stated in the box: for a fast driver acting through a slow medium (air temperature $\rightarrow$ heat flux $\rightarrow$ mixed layer) the two lengths add rather than compete — the flux pattern is laid down over the atmospheric correlation length and then carried away by the ocean — which is why the air-temperature row reads 1,889 to 3,833 km. The notation panel defines ℓ with a timeline. All parameter values are order-of-magnitude illustrations and can be changed live in the Dependency-Cone Explorer page.

Sources [Dependency-Cone Explorer](#) • [Chelton et al. 2011, global eddies](#)

Step 1 — each driver becomes a wedge in (lag, distance)

Each driver from the previous slide becomes a wedge on this chart. Horizontal axis: lag ℓ , how many months before now the input was observed (0 to 24). Vertical axis (logarithmic): how far from the pixel the input may sit and still matter, 10 km to 10,000 km.

A wedge rises to the right because older inputs have had more time to travel — reach grows as $v \cdot (\Delta t + \ell)$ — and stops abruptly where the driver's memory τ runs out. Wind is a tall spike at lag 0 only. Air temperature is a wide band for six lags. Surface temperature is a true wedge for eight lags. Interior temperature and salinity and sea-surface height are low and long. Own history is short.

The bold black line is the union of all wedges: the region a stencil for this target should follow. Its shape is an L: a spike at lag 0 (the atmosphere, wide but instant), a shoulder out to 6–8 months (mixed-layer temperature and integrated heat flux, 2,000–4,000 km), then a long low tail to 24 months (the interior, under 1,900 km).

Two reference lines: the dashed rectangle is our current "cylinder" — 89 stencil slots reaching 4,444 km at every one of 24 lags; the dotted line at 84 km is the 3-by-3 patch. The cylinder over-covers the tail by a factor of 2 to 50 and cannot cover the lag-0 spike at all.

Sources [Dependency-Cone Explorer](#)

Step 2 — the union becomes a per-channel, lag-dependent stencil

This slide turns the wedges into a stencil budget. Our production stencil, the "sunflower-89", spreads 89 slots over a disc of radius 4,444 km and is applied at all 24 lags: $89 \times 24 = 2,136$ "slot-lags" per channel — the cylinder.

The rule: slots needed for a given reach r scale with area, $89 \cdot (r/4,444 \text{ km})^2$, floored at 9 (the 3-by-3 patch) and capped at 89. Each row is a driver; each bar is a lag; bar height is the slots needed at that lag; grey stubs are lags outside the driver's memory, which are dropped.

Wind: one full-width sheet at lag 0, nothing else — 89 slot-lags. Air temperature: a wide sheet for six lags, 232. Surface temperature: a true cone from 9 to 44 slots over eight lags, 157. Interior temperature and salinity: a thin 24-lag column at the 9-slot floor for a year and a half before it widens, 238. Sea-surface height: a thin 12-lag column, 108 (its coastal arm is not counted by this isotropic formula). Own history: three short lags, 27.

Total: 851 cone-shaped slot-lags for the six channels against 12,816 for the cylinder — 93 % fewer. Equivalently, the saving could buy 48 months of history for the interior channels at no extra cost. These numbers follow directly from the illustrative speeds and memories; the Explorer page recomputes them if you change any parameter.

Sources [Dependency-Cone Explorer](#)

Step 3 — the physics behind each wedge

The physics behind each wedge, in four cards, and then what is fixed versus what is learned.

Ekman: wind stress τ drives a surface transport of about $\tau/(\rho f)$ at right angles to the wind — ρ is water density, f the Coriolis parameter set by latitude — within about one inertial period, roughly a day at 30° . Local, fast, shallow. At monthly cadence this is same-step forcing.

Thermal wind: the vertical shear of the geostrophic current is proportional to the horizontal density gradient, $\partial u/\partial z = (g/(f \rho_0)) \partial \rho/\partial y$. A gradient does not exist inside one pixel, which is why the 3-by-3 patch is the floor for temperature and salinity channels, and why heat flux matters through its spatial pattern rather than its local value.

Pressure gradient: the surface geostrophic current is $u = -(g/f) \partial \eta/\partial y$, where η is sea-surface height. SSH anomalies arrive slowly as westward Rossby waves in the interior and fast as Kelvin or coastal waves along boundaries.

Advection and eddies: momentum persists and mesoscale eddies drift westward at a few centimetres per second. The eddies themselves live four months and more, but at a fixed pixel the current decorrelates in weeks to a few months as they pass through — that is the memory used for "own history". Upstream cells carry what will arrive.

What the stencil fixes versus what attention learns: the cone is the support — which (cell, lag, channel) tokens may enter at all. It encodes only speed and memory per driver. Everything else — which upstream cell matters this month, how strongly an SSH gradient projects onto the current — is learned by the attention weights inside the support. A support too small cannot be learned around; one too large costs slots and admits noise.

Right box: the four ablations specialised to this target, with the predicted outcomes. They are hypotheses; nothing here has been run. The Dependency-Cone Explorer page uses the same formulas with editable parameters.

Sources [Dependency-Cone Explorer](#) · [Chelton et al. 2011, global eddies](#) · [Earth 2 paper \(v8, PDF\)](#) · [ml/CLAUDE.md \(replication rules, §3b\)](#)

Related work I — the dependency cone vs. the literature

Where does the dependency-cone idea stand against the literature? Row by row.

The origin is the Courant–Friedrichs–Lewy condition from 1928: a numerical scheme's stencil must contain the physical domain of dependence, $|\Delta x| \leq v \cdot \Delta t$. Our first rule is exactly this; the CFL condition has no notion of memory.

OceanNet (2024), a regional neural forecaster for the Gulf Stream, uses precisely a wave-speed-times-time argument to justify dropping lateral boundary conditions: Rossby waves at 40°N move about 1 cm/s, so eddies travel only 30–120 km from the open boundary over the forecast horizon. Same arithmetic, used to shrink a boundary rather than to shape a per-channel stencil.

PARADIS (2026) is the architectural version: a "neural semi-Lagrangian" layer gathers features at the departure point $x - \Delta t \cdot u$ — where the flow came from — turning the CFL limit into a smoothness condition on the velocity. A 20-parameter layer moves a tracer that a U-Net's fixed receptive field cannot. That is the tilted, anisotropic cone made operational, though with one transport field for all variables and no memory truncation.

ClimODE uses advection as an inductive bias but with fixed local and global branches. Limited-area weather models use fixed boundary halos, not sized by wind speed times lead time. Global models such as GraphCast, Pangu and FuXi set reach per step by hop count or window size and grow it with lead time by changing the step — the same reach for every channel. Per-variable tokenisation (ClimaX, Aurora, Prithvi WxC) gives each variable its own embedding, not its own spatial or temporal extent. Ocean emulators such as Samudra prescribe the atmosphere as forcing — our monthly-step inversion in practice.

What we did not find is the combination: truncating each process's cone by its own memory, taking the union per channel, and letting that dictate a channel-specific stencil. A one-day search is not a literature review; treat it as "not yet found", and cite OceanNet and PARADIS as the nearest neighbours.

Sources [Courant, Friedrichs & Lewy 1928](#) · [OceanNet](#) · [PARADIS — Learning to Advect](#) · [ClimODE](#) · [Neural-LAM](#) · [Stretched-grid ML weather](#) · [Keisler 2022](#) · [GraphCast](#) · [Pangu-Weather](#) · [FuXi](#) · [ClimaX](#) · [Aurora \(Nature 2025\)](#) · [Prithvi WxC](#) · [Samudra](#) · [ACE2-SOM](#)

Related work II — the four-sphere plan vs. the literature

The four-sphere ambition against the literature, by family.

Coupled emulators: Samudra is a 3-D global ocean emulator (temperature, salinity, currents at depth) trained on reanalysis; SamudrACE (2026) couples it to Ai2's machine-learned atmosphere ACE2. This is the closest structural analogue to our order — ocean interior first, atmosphere joined later — but it emulates a simulator rather than embedding observations, and has no biosphere. ACE2-SOM and ACE2-NEMO couple an ML atmosphere to a slab or dynamical ocean.

ECMWF's 2026 AIFS surface-ocean paper forecasts atmosphere plus sea-surface temperature, salinity, height, currents, waves and sea ice in one "component-agnostic" model, and explicitly gives the slowly evolving ocean and ice fields larger loss weights — the first ML system we found that names and handles the cross-sphere timescale mismatch. Surface only, and a forecast model rather than a reusable embedding.

Aurora (Nature 2025, titled "A foundation model for the Earth system") is one pretrained backbone with sphere-specific fine-tunes; atmosphere-centric, no interior ocean.

Observation-native learning: Aardvark Weather and GraphDOP learn directly from raw observations, skipping reanalysis; 4DVarNet learns to fuse sparse altimetry and temperature into gap-free surface fields. This is the recipe for taking Argo profiles, swaths and ship tracks natively — what "combine all relevant data" needs — but none of them has an Argo-depth encoder.

"Earth-system" foundation models: ESFM (2026) and Copernicus-FM unify heterogeneous sources with missingness tokens — the same primitive as our "masked" versus "never measured" tokens — but neither touches the ocean. The Prithvi family is a set of separate sphere-specific models; AlphaEarth uses ERA5-Land only as an input.

Ocean-biosphere and carbon ML (SOM-FFN, CSIR-ML6, CANYON-B, Granite) are task models with hand-chosen predictors. Coupled data assimilation (Penny & Hamill 2017) is where the timescale problem was first named. Platforms and twins (NVIDIA Earth-2, Destination Earth, EDITO) share the goal but are not learned embeddings — and NVIDIA's "Earth-2" is a name collision we must disambiguate, as is Aurora's title.

Sources [Samudra](#) · [SamudrACE](#) · [ACE2-SOM](#) · [ACE2-NEMO](#) · [AIFS surface ocean \(ECMWF 2026\)](#) · [Aurora \(Nature 2025\)](#) · [Aardvark Weather \(Nature 2025\)](#) · [GraphDOP](#) · [4DVarNet](#) · [ESFM](#) · [Copernicus-FM paper](#) · [Prithvi WxC](#) · [AlphaEarth Foundations paper](#) · [CSIR-ML6 \(Gregor et al. 2019\)](#) · [CANYON-B \(Sauzède et al. 2017\)](#) · [Granite-Geospatial-Ocean paper](#) · [Penny & Hamill 2017, coupled DA \(BAMS\)](#)

What should a generic Earth embedding see?

This slide opens the last part of the deck: if we want one embedding that is generic — usable for the ocean, the atmosphere, the land and both biospheres — what should it be allowed to see?

The rule at the top is the whole answer in one sentence: every signal that is relevant to the state of the four spheres at a point, and that cannot be derived from the other inputs. "Cannot be derived" is the important half. A vegetation index is a formula applied to two satellite bands; if the bands are in, the index adds nothing. Sunlight angle follows from position and time. The slope of a hill follows from the elevation map. None of those belong in the input — they are fine as things to predict, which costs nothing and gives the model free supervision.

The ladder on the left lists thirteen rungs, finest at the bottom. The bar is the native pixel size on a logarithmic scale, so 10-metre Sentinel imagery is a short bar and the 31-kilometre reanalysis is a long one. Colours are the sphere the rung mostly serves. Two rungs deserve a mention: rung 3, the 300-metre-to-1-kilometre daily radiometers, is the first rung that covers the ocean — the fine imagery of rungs 1 and 2 only sees land and coast. And rung 8 is the first rung that looks below the sea surface.

The box on the right is about derived products — a "reanalysis" is a weather or ocean model that is continuously nudged towards observations, and a "gap-filled map" is an observation product with the clouds interpolated away. They are admitted on three conditions: they must carry information we do not otherwise ingest (ERA5 has weather balloons and aircraft inside it), they are flagged so the model knows they are second-hand and are hidden from the input more often, and their time stamp is shifted by the window of observations they were built from, so they cannot smuggle in the future. That last point gets its own slide.

Sources [Generic-embedding input proposal \(full note\)](#) · [ERA5 documentation \(SST/sea ice prescribed; 12 h 4D-Var\)](#) · [DUACS system description \(windows\)](#) · [Lellouche et al. 2021, GLORYS12](#)

Input ladder I — fine and fast (rungs 0-4)

The first table walks up the fine end of the ladder.

Rung 0 is not data at all but the frame: where and when we are looking, and how deep. The model needs it because the same temperature means different things in January at sixty degrees north and in July at the equator.

Rung 1 is the 10-metre imagery: Sentinel-2, which is a colour-plus-infrared photograph of the sunlit surface with the atmosphere corrected out, and Sentinel-1, a radar that works day and night and through cloud, and that measures how rough and how wet a surface is. Between them they give field-scale texture — crops, forests, floods, sea-ice edges, ships. The constellation facts are current as of this week: Sentinel-2 has three satellites in orbit until the end of 2026, and Sentinel-1 is now the C and D satellites, since 1A was switched off at the end of June.

Rung 2 is Landsat, at 30 metres: the same kind of photograph plus a thermal image — how warm the ground is — and a forty-year archive.

Rung 3 is where the ocean appears. Sentinel-3 and VIIRS take a daily picture of the whole planet at 300 metres to a kilometre. Over the ocean the colour of the water tells you how much phytoplankton is in it — that is the ocean biosphere — and the thermal channels give the skin temperature of sea and land. MODIS, the older sensor everyone has used since 2000, is being retired at the end of this year, which is why VIIRS is named here.

Rung 4 is the geostationary satellites — the ones parked over one spot, like GOES over the Americas or Meteosat over Europe and Africa. They take a picture every ten minutes, so they are the only place where you can actually watch the atmosphere move. That is the fast end of the dependency cone.

The "O / D" column says whether each rung is observed or derived; the whole of this first table is observed.

Sources [Generic-embedding input proposal \(full note\)](#) · [Sentinel-2](#) · [Sentinel-2A extension to end-2026](#) · [Sentinel-1](#) · [Sentinel-1 orbital reconfiguration \(S1A end\)](#) · [Sentinel-1D data opening](#) · [Landsat Collection 2 Level-2](#) · [Sentinel-3](#) · [VIIRS](#) · [NASA MODIS → VIIRS transition](#) · [GOES ABI](#) · [GOES-19 operational](#) · [MTG FCI \(WMO OSCAR\)](#) · [Himawari AHI](#)

Input ladder II — fields and ocean memory (rungs 5-8)

The second table covers the physical fields and the ocean's memory.

Rung 5 is a collection of kilometre-scale sensors that are sparse — they see along a line or a narrow strip rather than everywhere. The important one for the ocean is altimetry: a radar that measures the height of the sea surface to a few centimetres along the satellite's track. Sea-surface height is the ocean's pressure map; its slopes drive the currents. SWOT is the new altimeter that sees a 2-kilometre-resolution strip instead of a line. The same rung holds ground weather radar, lightning, the laser altimeters that measure tree height and sea-ice thickness, and the satellites that measure greenhouse gases in the air column. Everything here is sparse in space or time, which is exactly the case where a set of points beats a grid — a theme that returns in the proposal.

Rung 6 is the cleaned-up daily maps: sea temperature with the clouds filled in, sea level everywhere and not only under the track, rain everywhere, wind over the whole ocean, sea-ice cover and drift, and the wetness of the top centimetres of soil. Most of these are derived — they merge several sensors and interpolate the gaps — so they are flagged, masked more heavily, and time-shifted. A few in this rung are genuine observations put on a grid, like the ASCAT scatterometer winds.

Rung 7 is the reanalysis atmosphere, ERA5: the best physically consistent estimate of the whole atmosphere at every hour since 1940, made by running a weather model and nudging it towards millions of observations. It is what forces everything else — heat and momentum into the ocean and land. One warning that surprised us in verification: ERA5's sea-surface temperature and sea ice are inputs to ERA5, taken from other products, not things ERA5 estimates, so they must never be treated as ocean observations. For real time, where ERA5 lags five days, the free ECMWF or GFS analyses at a quarter degree fill in.

Rung 8 is the first look below the surface: Argo floats, which drift at depth and surface every ten days to report temperature and salinity down to 2,000 metres — there are about 4,400 of them right now; deep floats to 6,000 metres; biogeochemical floats that add oxygen, nitrate, acidity and chlorophyll; moorings; surface drifters, which give currents directly; and GLORYS, the ocean reanalysis, a model's gridded best guess around all of these. This rung holds the heat capacity, the density gradients and the AMOC — the slow memory of the whole system. The note in the last column matters for the proposal: only points do this justice; gridding Argo throws away its depth resolution.

Sources [Generic-embedding input proposal \(full note\)](#) · [SWOT \(PO.DAAC\)](#) · [Sentinel-6](#) · [Jason-3 \(AVISO\)](#) · [MRMS](#) · [EUMETNET OPERA](#) · [GOES GLM](#) · [GEDI](#) · [ICESat-2](#) · [Sentinel-5P TROPOMI](#) · [OCO-2](#) · [OSTIA SST \(Copernicus Marine\)](#) · [MUR SST \(PO.DAAC\)](#) · [DUACS sea level L4 \(Copernicus Marine\)](#) · [GPM IMERG](#) · [ASCAT winds \(OSI SAF\)](#) · [AMSR2 sea ice \(NSIDC\)](#) · [OSI SAF sea-ice products](#) · [SMAP \(NSIDC\)](#) · [ERA5 \(CDS\)](#) · [ERA5 documentation \(SST/sea ice prescribed; 12 h 4D-Var\)](#) · [ECMWF open data](#) · [GFS \(NCEI\)](#) · [Argo](#) · [BGC-Argo](#) · [OceanOPS \(live platform counts\)](#) · [Global Drifter Program](#) · [GTMB / TAO moorings](#) · [OceanSITES](#) · [GLORYS12 \(Copernicus Marine\)](#) · [Copernicus global analysis/forecast](#)

Input ladder III — biospheres, truth, stage (rungs 9-13)

The third table completes the ladder with the two biospheres, the ground truth, and the stage.

Rung 9 is the ocean biosphere and carbon: ocean colour as observed, with the cloud gaps left as gaps; PACE, the new hyperspectral colour sensor launched in 2024; and the carbon measurements — SOCAT, which is forty-four million ship measurements of the CO₂ content of surface water, telling you whether the ocean is absorbing or releasing carbon at that spot; plus the machine-learned monthly carbon maps built on top of it, which are derived and flagged as such.

Rung 10 is the land biosphere and land water: leaf area, fluorescence — plants give off a faint glow when they photosynthesise, and a satellite can measure it — surface temperature, evaporation, soil moisture, snow, fire, the total water stored underground as measured by the GRACE gravity satellites, rivers, and the flux towers that measure carbon and water exchange every half hour at a few hundred sites. Almost everything here stays where it is: soil, plants and snow do not move sideways. That is the column-only cone family on slide 39.

Rung 11 is the ground truth: weather stations, balloons, aircraft, ships, buoys, tide gauges. These are the measurements the reanalysis of rung 7 is fitted to, so they are redundant while the reanalysis is present — and essential when it is masked out, because the model must then be able to rebuild the atmosphere from points.

Rung 12 is the static context: elevation, sea-floor depth, land cover, soil, tree height, glaciers, people. Depth sets what currents can do; terrain sets rain and rivers; soil sets what water does. Each is a single dot with zero speed and infinite memory.

Rung 13 is optional in the first phase: aerosols and greenhouse-gas columns, which matter for radiative forcing and close the carbon loop between land, ocean and air.

If one had to start with one rung per sphere, it would be rung 7 for the atmosphere, rungs 6 and 8 for ocean physics, rung 9 for the ocean biosphere, and rungs 1 and 10 for land — which is the sequencing agreed earlier: ocean interior first, then ocean colour, then the atmosphere as forcing, then land.

Sources [Generic-embedding input proposal \(full note\)](#) · [ESA OC-CCI](#) · [GlobColour L3 \(Copernicus Marine\)](#) · [PACE](#) · [SOCAT](#) · [Copernicus surface carbon](#) · [Copernicus Global Land LAI 300 m](#) · [TROPOSIF](#) · [MODIS LST \(MOD11A1\)](#) · [MODIS ET \(MOD16A2GF\)](#) · [GLEAM](#) · [H SAF \(ASCAT soil moisture\)](#) · [MODIS snow \(MOD10A1\)](#) · [IMS snow/ice](#) · [FIRMS active fire](#) · [GRACE/GRACE-FO JPL mascons](#) · [GRDC river discharge](#) · [GloFAS](#) · [Hydroweb.next](#) · [FLUXNET](#) · [ICOS](#) · [AmeriFlux](#) · [ISD stations](#) · [GHCN-Daily](#) · [IGRA radiosondes](#) · [ICOADS](#) · [NDBC buoys](#) · [GLOSS tide gauges](#) · [Copernicus DEM](#) · [GEBCO](#) · [ESA WorldCover](#) · [ESA CCI land cover](#) · [SoilGrids](#) · [ETH global canopy height](#) · [Meta/WRI 1 m canopy height](#) · [RGI glacier inventory](#) · [WorldPop](#) · [GHSL](#) · [VIIRS Black Marble](#) · [MAIAC AOD \(MCD19A2\)](#) · [MERRA-2](#)

Observed, derived, or derivable — and the leakage trap

This slide sorts every input into three bins and then explains a trap that is easy to fall into.

Left, always in: observations — a number a sensor produced. A reflectance, a radar echo, a float's thermometer reading. Nothing else contains them, so they are irreducible.

Middle, admitted but flagged: derived products — a number a program produced from observations plus a model or a statistical assumption. Reanalyses, gap-filled maps, machine-learned reconstructions. They come in because they carry information we do not ingest raw, but every dot carries a "source token" saying what kind of thing it is, and derived sources are removed from the input at least half the time. That way the model cannot get lazy and simply copy the reanalysis; it must also be able to work from observations alone.

Right, dropped: things that are pure functions of inputs already present — vegetation indices, sun angle, slope, the currents you can compute from the sea-level slope. They are not inputs. They are optional prediction targets.

The orange band is the trap. A gap-filled map or a reanalysis for a given day is built from observations on both sides of that day. The delayed-time sea-level maps, for example, use satellite tracks from up to six weeks before and after the day in question — only the near-real-time stream is built from the past alone; the ocean reanalysis works in seven-day chunks; ERA5 in twelve-hour chunks. If we train the codec to predict the present or the future from the past, then a derived dot labelled "ten days ago" secretly contains information from a few days ago — or even from after the anchor time. The model would learn to peek. The rule is simple: shift each derived source's effective time by its window, never use a delayed-time product at lag zero, and train the forecasting part of the objective on the near-real-time streams, which by construction only saw the past. This gives two embeddings per anchor — a near-real-time one and a final one — exactly the way ECMWF publishes ERA5T first and ERA5 later.

Sources [Generic-embedding input proposal \(full note\)](#) · [DUACS system description \(windows\)](#) · [Pujol et al. 2016, DUACS DT2014](#) · [Lellouche et al. 2021, GLORYS12](#) · [ERA5 documentation \(SST/sea ice prescribed; 12 h 4D-Var\)](#) · [OSTIA SST \(Copernicus Marine\)](#) · [GlobColour L3 \(Copernicus Marine\)](#) · [Dynamic World](#) · [MODIS NDVI/EVI \(MOD13Q1\)](#) · [OSCAR currents](#)

Five cone families — one shape per rung

Now we apply the dependency cone from the earlier slides to every rung of the ladder. The recap: a process that travels at speed v can only influence a point from within a distance v times the time elapsed, and only lags shorter than the process's memory τ are worth reading. When we do that for all thirteen rungs, five shapes come out.

Family A, fast, wide and short, is the atmosphere: wind carries information at about ten metres per second, which is 860 kilometres a day, but the atmosphere forgets in three to ten days. So the support is a wide flat disc — you need dots from far away but only from the last few days.

Family B, slow, narrow and long, is the ocean interior and sea level: currents move at a tenth of a metre per second, 13 kilometres a day; sea-level anomalies drift westward as Rossby waves at 3 kilometres a day. But the memory is months for eddies and years to decades for the interior. The support is a thin tall column, tilted upstream.

Family C is the L-shaped union we met before: sea-surface temperature and salinity, sea ice, surface chlorophyll. They are pushed around by the fast atmosphere and remembered by the slow ocean, so the support is wide at short lags and narrow at long lags.

Family D is land. Soil, plants and snow do not move sideways at all; they are forced from above by the atmosphere and remember locally for weeks to seasons. The support is a narrow column with a wide flat hat on top — the atmosphere disc at the shortest lags. Rivers and sea ice are the exceptions that move along a path.

Family E is static: one dot, zero speed, infinite memory.

The paragraph at the bottom adds depth as a sixth axis. The top ten to a hundred metres of the ocean, the mixed layer, are stirred within hours and belong to the surface cone; below it memory grows with depth — months at a few hundred metres, decades below two thousand. So depth dots are spaced logarithmically, dense near the surface, each depth band with its own memory. The consequence for sampling is the point of the slide: fine rungs want a small dense patch and a long local history; coarse physical rungs want a large sparse far field and a short history; the interior wants depth and years. No single grid and window serves all three. A per-channel dot sampler does.

Sources [Generic-embedding input proposal \(full note\)](#) · [Dependency-Cone Explorer](#) · [Chelton et al. 2011, global eddies](#) · [Deser, Alexander & Timlin 2003 \(SST persistence\)](#) · [Zhang et al. 2019, predictability limit](#)

Proposal — a cone-native codec: dots in, embedding out

This is the proposal itself: a cone-native codec — dots in, embedding out.

Until now the codec embedded one pixel-month, and a second stage assembled a stencil of those embeddings across space and time. Here the codec itself reads a set of dots — each dot is a sample of one channel at some offset in space, some depth, and some lag into the past — drawn from that channel's cone around an anchor point. The picture on the left shows it: the anchor is the yellow dot at the origin; the orange wind dots spread wide but stay near the present; the blue current and sea-level dots stay close in space but reach far back in time; the teal temperature dots fill the L. Depth dots are the small stack at the anchor — a float profile enters as it is, not regridded.

Step 1, the sampler, is fixed by physics. For gridded channels we draw dots in log-polar space and log lag, denser near the anchor and near the present, with the slot budget from the family. For point, profile and track channels — Argo, drifters, stations, altimeter tracks — the dots are the actual observations that fall inside the cone. Nothing is interpolated or invented. Fine imagery arrives as tokens from a small local codec, which the hybrid slide explains. Missing dots are simply absent: a set has no holes to fill.

Step 2 turns each dot into a token: the value, the coordinates as Fourier features on a log scale so that ten metres and ten thousand kilometres fit in the same code, a channel embedding, and the source flag from the previous slide.

Step 3 is the encoder, Perceiver-style: a couple of hundred learned latent vectors cross-attend to the two to four thousand dots, then a few self-attention blocks over the latents. The cost grows linearly with the number of dots, and per anchor it is a few billion operations — less than running a standard vision transformer on one image.

Step 4 pools the latents into the dynamic embedding, 256 numbers stored as 8-bit integers.

Step 5 is the objective. A held-out dot becomes a query — coordinates and channel without the value — and a light decoder predicts its value as a distribution. We mix several ways of hiding dots: whole channels, as before; the recent past, so that the model must forecast from the older past; dots from the future, so that the embedding learns tendency; an upstream sector; a depth band; and always the anchor's own channels, so the embedding still means "the state at this point". The distinction between "never measured" and "hidden" survives on the target side: there is no loss where nothing was ever observed.

Sources [Generic-embedding input proposal \(full note\)](#) · [Perceiver IO](#) · [AIFS surface ocean \(ECMWF 2026\)](#) · [Earth 2 paper \(v8, PDF\)](#)

Where velocity comes from — two dots of the same field at two lags

This slide answers the question in the request directly: can an embedding capture momentum features — velocity, water flow — if it sees several points in space and time? Yes, and here is why.

A single photograph does not show speed. Two photographs a known time apart do: whatever moved, moved by speed times time. In the diagram, position runs left to right and lag runs upward, the present at the bottom. A feature carried along by the flow — a warm patch, an eddy, a cloud — traces a tilted band. The yellow dot is the anchor now; the blue dot is the same water some time ago, upstream by exactly speed times lag. The grey dots are candidates at the same lag that the model has to reject.

The argument in prose: if a field is carried along, its value here now is best predicted by its value upstream a while ago. The correlation between the anchor's neighbourhood now and the field some time ago peaks at a displacement equal to speed times lag. Oceanographers have used exactly this — the maximum cross-correlation method — since the 1980s to compute currents from pairs of satellite temperature images. Attention over dots that carry their relative position and lag can implement that search, and the winning ratio of displacement to lag is a velocity. The rate of change at one place comes from two lags at that place; convergence comes from dots around the anchor. A per-pixel snapshot codec has one lag and cannot do any of this.

Having the capacity is not the same as using it, so three measures make sure it is used. First, flow enters as maskable channels — drifter velocities, sea-ice drift, wind, river discharge, and the flagged reanalysis currents — so the masked objective directly asks the embedding to produce velocity whenever the velocity channel is hidden. Second, the slow families always contain several lags of the same field, by construction of the cone budgets. Third, optionally, a fraction of the dots can be drawn along the path a first-guess velocity would predict — the semi-Lagrangian idea from the PARADIS paper used as a sampler rather than as a layer.

The test is a velocity probe: a simple linear head from the frozen embedding to drifter velocity or reanalysis currents at the anchor, with the derived channels hidden at test time. The snapshot embedding should score near zero; the cone codec should not. It is the cheapest decisive experiment in the plan.

Sources [Generic-embedding input proposal \(full note\)](#) · [Emery et al. 1986, maximum cross-correlation \(JGR\)](#) · [PARADIS — Learning to Advect](#) · [Global Drifter Program](#) · [GLORYS12 \(Copernicus Marine\)](#)

Is it zero-sum? Thin codec + thick stage 2 vs cone codec + thin stage 2

This is the slide the request asked for explicitly: is the architecture zero-sum? Either we keep the embedding fine-grained and local and let stage 2 make the spatial and temporal connections, or we put that work into the embedding — and does one side's gain equal the other's loss?

Column A is the thin codec with a thick stage 2: embed one pixel-month, then a second-stage model assembles stencils of those embeddings. That is today's Earth 2, and it is what every model in the survey does — they embed images, and the user connects them. Column B is the cone codec from the previous slides with a thin stage 2.

Reading down the rows, green is an advantage and red a cost. A wins on cost per embedding, on caching — an embedding is this place at this time and never changes — on provenance, on simplicity, and on direct comparability with published models. B wins on everything that involves time: dynamics, which a snapshot cannot represent at all; the information bottleneck, because a per-pixel codec has to compress before it can know what the dynamics need, and whatever it throws away as noise — the tiny sub-pixel shifts and gradients that carry velocity — is gone for good; sparse observations, because a lone Argo profile with no context is nearly empty but a profile among its neighbours is not; the cone geometry, which A has to re-learn for every task from labels and B fixes once from physics; sample efficiency of the probes, which matters when the AMOC has only twenty years of monthly labels; and forecasting, which B gets for free inside pretraining.

Two rows are honest costs of B: it is a more complex training pipeline, and there is a real risk that the model copies the derived channels instead of learning — which is what the source flags and drop rates are for. And for purely static tasks like land cover, A's snapshot semantics are actually better.

So the table is not one-sided, but it is also not symmetric — which is the argument of the next slide.

Sources [Generic-embedding input proposal \(full note\)](#) · [Earth 2 paper \(v8, PDF\)](#) · [ml/C LAUDE.md \(replication rules, §3b\)](#)

Not zero-sum — the asymmetry, and the split that follows from it

The answer to "is it zero-sum" is no, and the reason is an asymmetry.

First: B contains A. A cone codec that ignores every dot except the anchor's own is a per-pixel codec. Whatever A can represent, B can. The reverse is false, because of the bottleneck: information that the per-pixel code discards before the dynamics are visible cannot be recovered by stage 2 — this is the data-processing inequality, the formal statement that you cannot get information back by processing. So in capability the trade is one-sided; what B pays is compute and a less tidy meaning of "the embedding of this place".

Second: the compute is paid in different places. A pays little per embedding but pays the context assembly again inside every stage-2 model, for every task. B pays once per anchor and amortises it across tasks — provided the cone is generic. And that is exactly what a physics-set support promises: the shape depends on the channel, not on the task.

That gives a principled split rather than a compromise. Into the embedding goes everything whose relevance is decided by physics — the cone, depth coupling, the local past — because that is identical for every task. Into stage 2 goes everything whose relevance is decided by the task: which section to integrate for the AMOC, which basin to sum for carbon, which teleconnection matters — El Niño's Pacific state is ten thousand kilometres and six months away, outside any single cone — and which label.

The right side shows the resulting hierarchy. Raw observations at the bottom. A local codec, per tile and date, cached, with snapshot semantics — the existing per-pixel codec generalised to a tile — producing a small local code that mapping tasks use directly. A cone codec per anchor, which reads local tokens plus coarse dots plus native point observations inside each channel's cone, and produces the dynamic code where the momentum features live. And a thin stage 2 for teleconnections, aggregation and task heads.

What does existing evidence say? Nothing decisive yet, and it is worth being precise about why. The attribution matrix's contrast between a 3-by-3 codec and the same 3-by-3 raw input — 0.672 against 0.659 — is a single-seed probe number, and our own record shows probe numbers moving by far more than that between seeds, so the honest reading is parity. And on 2 September the paper was reset: every rolled forecast number from before that date was withdrawn, because the second-stage heads had been trained under a pool that had seen the held-out years. The first clean measurement of the bottleneck question is the one the reset paper puts first on its list — a Linear Inverse Model fitted in pixel space against one fitted in the codec's embedding space. If the embedding-space model is worse, the codec has discarded predictable signal, which is exactly column A's weakness. Phase 0, next slide, is built to run after that measurement exists.

Sources [Generic-embedding input proposal \(full note\)](#) · [Earth 2 paper \(v8, PDF\)](#) · [ml/CRAUDE.md \(replication rules, §3b\)](#) · [Prithvi-EO 2.0 paper](#)

Two stencils, one cone — built, run, and re-cut for v2

"Two stencils, one cone" is the design that splits the DEPENDENCY CONE — everything in space and time that could have influenced one pixel — between the two models we train. The CODEC is the compressor: it reads raw channel values sampled as dots around one anchor pixel over the last six PENTADS (a pentad is a fixed five-day bin, so six of them is thirty days) and squeezes them into a 32-number embedding. STAGE 2 is the forecaster: it reads a cylinder of those embeddings reaching two years back. The picture on the left plots lag along the bottom and distance from the anchor up the side, both on logarithmic scales. Two identities are asserted on that grid and checked by a coverage test in the code: the two stencils together cover the whole cone, and they overlap in exactly one place — the anchor's own column, which is why the yellow bar along the bottom is drawn separately.

The darker shapes are the geometry as built, sized at 0.3 metres per second: the codec owns a disc growing to 907 kilometres over six lags, stage 2 owns everything beyond it out to a 4,444-kilometre cap. The lighter dashed shapes are cone v2, sized at 5.4 metres per second — the fastest measured surface current on Earth times a 1.5 safety factor. At that speed the codec's own thirty-day window already reaches 16,330 kilometres, most of the planet, and stage 2's stencil is a global band at every lag. So the division of labour stops being near field against far field and becomes recent against old: thirty days of raw channels against two years of embeddings, both of them global. Every v2 number on this slide is a DESIGN value. Nothing about v2 has been built, trained or measured.

The right-hand box is what HAS been measured. Three hypotheses were registered with their falsifiers before anything ran. H1: the cone codec's embedding carries the anchor's own water velocity better than a codec that sees only one snapshot. H2: a forecast head built on cone embeddings beats the control at five to thirty days. H3: the thin outer stencil does as well as the full cylinder it replaces. H1 is the one with an answer, and the answer is REFUTED — which here means the pre-registered failure branch actually happened, at five seeds that agree, not that the measurement was inconclusive. The read-out is a ridge regression from the frozen 32-number embedding to the current at the anchor, with the current channels hidden from the encoder so it cannot simply copy them. Its control is the RAW-PATCH BAR: the same ridge fitted directly to today's three-by-three patch of sea-surface height, with no codec involved at all — geostrophy read straight off the pixels. Thirty days of history landed at or below that bar at every seed. What the history did buy is persistence — a hidden sea-surface-height or temperature channel reconstructed far better than from today's patch alone — and a seven-to-eight-per-cent better forecast one and two pentads ahead. That, not velocity, is what cone v2's case now rests on.

Sources [E-069 plan — two stencils, one cone](#) · [E-069 in the experiment log](#) · [E-071 plan — cone v2: global, harmonic, Argo-aware, sized by the fastest thing in the ocean](#) · [Generic-embedding input proposal \(full note\)](#) · [Chelton et al. 2011, global eddies](#) · [Deser, Alexander & Timlin 2003 \(SST persistence\)](#) · [Emery et al. 1986, maximum cross-correlation \(JGR\)](#) · [Perceiver IO](#) · [GLORYS12 \(Copernicus Marine\)](#) · [ml/CLAUDE.md \(replication rules, §3b\)](#)

What feeds the cone codec today — four products, 42 channels, one axis

This slide is the inventory: everything the codec being trained right now is actually allowed to see, and everything it is not.

Four columns of vocabulary first. A REANALYSIS is a computer model of the ocean or the atmosphere that is repeatedly nudged towards whatever measurements exist, so that it produces a complete, gap-free history — GLORYS12 for the ocean and NCEP Reanalysis 1 for the wind are both of that kind, which is why they are marked red for "derived": they contain a model's opinion as well as observations. NOAA's sea-surface-temperature analysis and the Roemmich-Gilson Argo product are marked green because they only interpolate measurements; no ocean model is involved. A PENTAD is a fixed five-day bin, which is the clock of this tensor: forty-three years of them, 1982 to 2024. And an ASSIMILATION WINDOW is the span of observations a reanalysis is allowed to look at around each day — GLORYS uses seven days, so a GLORYS value dated Monday already knows something about Thursday. Five of the forty-two channels have that problem and need their effective time shifted.

The graph reads top to bottom: instruments, then the four products we download, then the channels those products become, then what training computes for itself. Two things in the bottom row are worth stating carefully. An ANOMALY here means a channel with its own average for that calendar month subtracted, and the average is built from the TRAINING years only; a HOLDOUT year — 2009, 2017, 2023 — is a year deliberately withheld so that skill can be measured on data the model never saw. That is exactly why the app's published 1991–2020 normal must not be reused as the baseline: it averages over the held-out years, and using it would quietly leak the answer into the training set. And the RAPID and Florida Current transports are LABELS, not inputs: they sit in the same file, they are what the probes are scored against, and the encoder never reads them. An input is something the model sees; a label is something it is asked about.

The panel on the right is the honest list of what is missing, ranked. The ordering rule is information per byte and non-derivability: heat and freshwater forcing first because there is no forcing of that kind in the tensor at all, then depth, then the observed sea surface the reanalysis was fitted to. The biosphere is last as an input and first as a target — the data ladder rejected chlorophyll as an input for the overturning question because there is no mechanism at these timescales, which leaves it as the ideal held-out thing to predict.

Sources [GLORYS12V1 product page \(Copernicus Marine\)](#) · [NOAA OISST v2.1 \(NOAA PSL\)](#) · [Roemmich-Gilson Argo climatology \(Scripps\)](#) · [NCEP/NCAR Reanalysis 1 \(NOAA PSL\)](#) · [RAPID-MOCHA-WBTS 26.5° N data](#) · [Florida Current cable transport \(NOAA AOML\)](#) · [ml/plans/DATA_LADDER.md \(source\)](#) · [ml/plans/E069_cone_codec.md \(source\)](#) · [E-069 plan — two stencils, one cone](#) · [CONE_DATA_AND_ENSO.md — the companion page](#) · [Generic-embedding input proposal \(full note\)](#) · [ml/CLAUDE.md \(replication rules, §3b\)](#)

Boundaries — four edges, four rules, and two of them now change

Every model has edges. This slide is about the four the cone codec actually meets, what it does at each, and the two that Chris decided on 4 September to remove altogether.

Some words first. A PENTAD is a five-day bin — the time step of the tensor, counted from the first of January 1982, so the whole archive is 3,142 of them. An ANOMALY is what is left of a measurement once the normal value for that time of year is subtracted; that normal is the CLIMATOLOGY, and ours is built from training years only so the test years cannot leak into it. The ATTENTION MASK is the list of tokens the transformer is allowed to look at at all. And the codec has two different ways of saying "no value here", which is what the whole left half of the slide turns on. An INVALID token is a position that does not exist — the dot fell outside our rectangle — and it is deleted from attention. A MISS token is a real position where the world simply was not observed, land under a sea-surface sensor for instance, and the model does learn from it, because "nobody measured this" is itself information.

Panel A is the geographic edge, and it is the first of the two that goes. The tensor is a North Atlantic window, so a dot falling off the eastern side is clipped and thrown away rather than wrapped round to Florida; twenty-two per cent of anchors lose at least one dot that way and 2.68 per cent of all dots are lost. The decision, in green, is that the dataset becomes GLOBAL: a 721-by-1,440 grid running from pole to pole, longitude wrapping arithmetically, and each dot computed as a DESTINATION POINT on the sphere — start where the anchor is, travel a given distance on a given bearing, and read off where you land — so a dot that would cross the pole simply comes out on the far side, with no special case. Those lost dots become real dots and the truncated cones become full ones.

Panel B is land: another eight per cent of dots, marked missing rather than invalid, and that rule was right and stays. Panel C is time — an anchor whose cone runs off either end of the archive is refused outright, and a held-out year blocks eight pentads of anchors around itself so no training anchor can peek into it, at a cost of about one per cent of the training bins. Both rules stay, and the grey tag on each panel says so.

Panel D is the edge we introduced ourselves, and it is the second one that goes. Chopping the year into twelve calendar boxes means 411 of 3,142 pentads straddle a boundary and 106 of them have just one of their five days inside the month whose average is subtracted — a sawtooth in the baseline at every month end. The decision is a DAY-OF-YEAR HARMONIC climatology: instead of twelve separate box averages, fit one smooth curve through the year made of a constant plus a few sine and cosine waves — one cycle per year, two, three — and evaluate it at each bin's middle day. Seven numbers replace twelve, the baseline is continuous across every month and every year end, and Argo's 252 live pentads across twenty-one years are fitted together instead of about twenty-one per box. Both decisions are specified in the E-071 plan and neither is implemented; nothing on this slide about them is measured.

Sources [ml/cone_sampler.py — the sampler, its off-grid rule and the holdout certificate \(source\)](#) · [ml/train_cone.py — anchor drawing and --holdout-scope window \(source\)](#) · [ml/trainprobe.py — the anomaly transform \(source\)](#) · [ml/build_family4.py — the pentad tensor builder \(source\)](#) · [E-034 plan — the pentad tensor and the Argo missing-token decision](#) · [E-059 plan — --holdout-scope window, why the pool discipline exists](#) · [E-069 plan — two stencils, one cone](#) · [ml/plans/E069_cone_codec.md \(source\)](#) · [E-071 plan — cone v2: global, harmonic, Argo-aware, sized by the fastest thing in the ocean](#) · [Argo](#) · [Roemmich-Gilson Argo climatology \(Scripps\)](#) · [CONE DATA AND ENSO.md — the companion page](#) · [ml/CLAUDE.md \(replication rules, §3b\)](#)

The speeds the cone did not have — and the speeds it will

Chris asked why the overturning circulation's own speed appears nowhere in the model, and then decided how to fix it: size every channel family from the MAXIMUM measured speed of the fastest mechanism that carries it, times 1.5.

Words first. ADVECTION is transport by the water itself moving — a warm parcel physically carried north. WAVE PROPAGATION is different: a disturbance travels through the water while the water largely stays put, so a signal crosses a basin far faster than any parcel. A KELVIN WAVE is the fast kind trapped against a coastline or the equator. BAROCLINIC means the motion is felt mainly in the density layering of the interior and travels at metres per second; BAROTROPIC means the whole water column moves together, which is far faster — about two hundred metres per second — because it is essentially a shallow-water gravity wave. The ANTIPODE is the point on the opposite side of the planet, 20,015 kilometres away by the shortest surface path, which is the farthest anything can be from an anchor and therefore the natural cap on any reach. An OCTAVE of distance is a doubling of it; LOG-RADIAL placement means putting the same number of dots in each doubling, so the near field stays as densely sampled when the cone gets eighteen times wider. A PERCEIVER is the encoder we use: a small fixed set of learned vectors reads over all the input tokens, so cost grows in a straight line with the number of tokens rather than with their square. A PROFILE TOKEN is one token carrying a whole vertical column — sixteen temperatures and sixteen salinities down the pressure ladder — instead of thirty-two separate tokens for the same water.

Every number on the ruler is a maximum, and each one is literature or our own data, named beside it. The Somali Current's seven knots, about 3.6 metres per second, and the Agulhas core's 2.45 come from Wikipedia's articles on those currents; the Gulf Stream's roughly 2.5 from NOAA's Ocean Service page; the first-baroclinic Kelvin wave's 2.8, and the barotropic 200, from Wikipedia's Kelvin wave article, with the phase-speed atlas of Chelton and colleagues from 1998 behind the baroclinic number; the long Rossby wave's 0.03 from Chelton and colleagues 2011, already cited in our own plan; the deep western boundary current's 0.02 to 0.1 is standard oceanography. Two numbers are ours: 1.62 metres per second for the Kuroshio off Taiwan and 2.79 for the Somali Current in July 2020, both computed from our own GLORYS bake in data/currents_y, which is a monthly one-degree average and therefore a lower bound on what the real ocean does.

The design speed is the largest of those, 3.6, times 1.5, giving 5.4 metres per second, or 2,333 kilometres per five-day bin. The buffer covers three things: the tensor is a five-day mean on a quarter-degree grid and under-reads peaks, the maxima are published values rather than true extremes, and a warming ocean may run faster. It is spent on the wide side deliberately — a cone that is too wide only costs tokens, while a cone that is too narrow silently asserts that a real cause cannot have arrived. Wind is beyond any cone at all: the jet stream crosses the planet inside one bin, so that family gets a ring of dots out to the antipode instead. Argo stops being a zero-reach vertical column and becomes twenty-six profile tokens at the two lags where the product is actually live. None of this is measured, and after E-069's velocity result the case for it is rolled forecast skill, not velocity.

Sources [ml/cone.py — the cone geometry and the channel families \(source\)](#) · [ml/cone_sampler.py — the sampler, its off-grid rule and the holdout certificate \(source\)](#) · [E-069 plan — two stencils, one cone](#) · [E-069 in the experiment log](#) · [ml/plans/E069_cone_codec.md \(source\)](#) · [E-071 plan — cone v2: global, harmonic, Argo-aware, sized by the fastest thing in the ocean](#) · [Somali Current \(Wikipedia\)](#) — up to 7 knots $\approx$ 3.6 m/s · [How fast is the Gulf Stream? \(NOAA Ocean Service\)](#) — about 9 km/h $\approx$ 2.5 m/s · [Agulhas Current \(Wikipedia\)](#) — core 2.45 m/s · [Kelvin wave \(Wikipedia\)](#) — first-baroclinic $\approx$ 2.8 m/s; barotropic $\approx$ 200 m/s · [Chelton et al. 1998 — first-baroclinic Rossby radius and gravity-wave phase speed atlas \(OSU\)](#) · [our own GLORYS surface-current bake, monthly 1° \(data/currents_y\)](#) · [Generic-embedding input proposal \(full note\)](#) · [Chelton et al. 2011, global eddies](#) · [GLORYS12 \(Copernicus Marine\)](#) · [Argo](#) · [Global Drifter Program](#) · [RAPID-MOCHA-WBTS 26.5° N data](#) · [Florida Current cable transport \(NOAA AOML\)](#) · [ml/plans/DATA_LADDER.md \(source\)](#) · [CONE_DATA_AND_ENSO.md — the companion page](#) · [ml/CLAUDE.md \(replication rules, §3b\)](#)

Land, ice and air — nothing is dark by design

Chris asked why there is no land data at all — "it can be sparse for now, but still better than absolute darkness" — and, separately, that some channels be available over both the ocean and the land. This slide is the answer, and it turns on one definition.

A SHARED CHANNEL is a channel defined by WHAT is measured and by WHICH INSTRUMENT, never by which sphere it sits over. Today all forty-two channels are ocean channels, so every land cell is told forty-two times per five-day bin that the world was not observed there — which is false; it was observed, by instruments we already list. The fix is a SPHERE CODE: a per-cell label saying ocean, land, ice sheet or inland water, read by the codec as a coordinate exactly the way depth already is. One temperature channel then means sea-surface temperature at one anchor and land-surface temperature at the next.

Vocabulary, first half. SKIN temperature is the surface itself as an infrared sensor sees it — the top fraction of a millimetre of water, or the ground and the leaves; 2 M temperature is the air two metres above it, what a thermometer in a weather screen reads. The two differ by several degrees on a sunny afternoon, which is why both are channels. SIGMA-NOUGHT is the radar cross-section, how much of a radar pulse the surface sends back: over water it rises with wind and waves, over land with soil moisture and vegetation — the same instrument, ASCAT, looking at both. ALBEDO is the fraction of sunlight reflected: near zero for open ocean, up to 0.8 for fresh snow.

Vocabulary, second half. BRIGHTNESS TEMPERATURE is the natural microwave glow of a surface, as a passive radiometer measures it, written as the temperature a perfect emitter would need to glow that brightly. SMAP's L-band version becomes sea-surface salinity over water and soil moisture over land — so the raw brightness temperature is the shared channel and the two products become things to predict. GRACE is a pair of satellites that weigh the water, ice and groundwater beneath them through tiny changes in gravity. ALTIMETRY is a radar measuring a surface's height from orbit: over an ice sheet, the same technique measures how fast the surface is dropping.

A PROFILE TOKEN carries a whole vertical column in one token instead of one per level, and the same type serves an Argo temperature-and-salinity profile, a stack of soil layers, and the firn — the compacted old snow on top of an ice sheet. Two design points go with it. LAND DOES NOT ADVECT: soil, snow and vegetation do not travel sideways, so their cone has speed zero and its reach is only the distance over which the field stays correlated, a few hundred kilometres. And ERA5-LAND is a higher-resolution land rerun of ERA5 with no data assimilation of its own; plain ERA5 is the assimilating one, which is why ERA5 is the source in the table.

"Sparse for now" is Phase L0: six gap-free channels from two sources, covering the whole 1982-onwards axis and the whole globe including Antarctica. Everything that only starts in 2000 waits for L1. Nothing on this slide is built or measured. The dataset specifications reused here — resolutions, cadences and record starts — come from the proposal's verification pass of 2 September; the ones added for this slide (MODIS albedo, ITS_LIVE, CryoSat-2, ESA CCI soil moisture, IGRA) have NOT been through it and must be verified before anything is fetched.

Sources [E-071 plan — cone v2: global, harmonic, Argo-aware, sized by the fastest thing in the ocean](#) · [Generic-embedding input proposal \(full note\)](#) · [ml/plans/DATA_LADDER.md \(source\)](#) · [ERA5 \(CDS\)](#) · [ERA5 documentation \(SST/sea ice prescribed: 12 h 4D-Var\)](#) · [MODIS LST \(MOD11A1\)](#) · [MODIS snow \(MOD10A1\)](#) · [MODIS BRDF/albedo \(MCD43A3, 500 m daily\)](#) · [ASCAT winds \(OSI SAF\)](#) · [SMAP \(NSIDC\)](#) · [GRACE/GRACE-FO JPL mascons](#) · [ICESat-2](#) · [CryoSat-2 \(ESA\) — ice-sheet elevation change](#) · [ITS_LIVE — glacier and ice-sheet surface velocity](#) · [OSI SAF sea-ice products](#) · [ESA CCI soil moisture](#) · [IGRA radiosondes](#) · [GPM IMERG](#) · [OSTIA SST \(Copernicus Marine\)](#) · [DUACS sea level L4 \(Copernicus Marine\)](#) · [Hydroweb.next](#) · [GLEAM](#) · [GLORYS12 \(Copernicus Marine\)](#) · [Argo](#) · [CONE_DATA_AND_ENSO.md — the companion page](#)

The Earth foundation model — the data decides the size

Chris asked for an Earth foundation model of one to ten billion parameters, possibly a mixture of experts, with a Chinchilla scaling ladder and a comprehensive study of how to design the training data. This slide answers the last part, because it turns out to decide the first.

CHINCHILLA is the 2022 DeepMind scaling result: for a fixed compute budget a model is best trained at roughly twenty training tokens per parameter, and the models before it were far too big for the data they saw. Turned around, twenty times the parameter count is the amount of data a model of that size can be justified in having. Two words then need separating. A VALUE is one measured number — one channel, at one cell, in one five-day bin. A TOKEN in this design is one place at one time carrying the whole value vector of every channel there, so with sixty-five channels one token is about sixty-five values; that is why the token column is the value column divided by sixty-five. The CEILING column is values divided by twenty: the largest model the data could support if every value were independent.

That last clause is why the column says ceiling and not target, and the orange box is the correction. Neighbouring quarter-degree cells are not independent — they are correlated over roughly three hundred kilometres, and at the surface over two to six months. In the North Atlantic that is about a hundred and fifty cells and fifteen five-day bins per genuinely INDEPENDENT SAMPLE, so two and a half billion raw values are worth of order one to ten million independent ones. Which is exactly the regime in which our measured width ladder is flat: a 7.6-million-parameter head and a 206-million-parameter head roll the same.

Which numbers are MEASURED in this programme and which are the plan's stated ASSUMPTIONS? Measured: the flat width ladder (0.103 / 0.104 / 0.105), the LIM result — a linear inverse model is a plain linear forecast fitted to the same field, and ours beats every learned head from fifteen days out — the held-out minima inside two thousand steps, and Argo being eighty per cent of the bytes for about one per cent of the information. Assumed: every value count in the ladder, the sixty-five-channel token, the divide-by-twenty anchor, the correlation lengths and the $\times 1.5$ discount on daily cadence. Nothing here is dispatched.

Sources [E-072 plan — the Earth foundation model: data design first, then the scaling ladder, then the recipe](#) · [Handover, 4 Sep 2026 — cone v2, the four spheres, the Earth foundation model, and where to fork](#) · [ml/SCALING.md — how skill moves with model and data size \(the programme's scaling audit\)](#) · [Hoffmann et al. 2022 — Chinchilla: training compute-optimal large language models \(~20 tokens per parameter\)](#) · [Muennighoff et al. 2023 — scaling data-constrained language models \(the \$\leq 4\$ -epoch limit\)](#) · [E-070 plan — the global tensor \(family 7\): the globe at 0.25° pentads, gates G1–G3, phases A–F](#) · [E-071 plan — cone v2: global, harmonic, Argo-aware, sized by the fastest thing in the ocean](#) · [ml/plans/DATA_LADDER.md \(source\)](#) · [E-062 in the experiment log — the flat width ladder and the Argo information finding](#) · [E-066 in the experiment log — the 200-mode linear inverse model as the reference forecast](#) · [E-069 plan — two stencils, one cone](#) · [Generic-embedding input proposal \(full note\)](#) · [Earth 2 paper \(v8, PDF\)](#) · [ml/CLAUDE.md \(replication rules, §3b\)](#)

The scaling ladder and the recipe

The left half is the ladder of things to train, in order. The right half is how to train them. Six rungs, L0 to L5. L0 is done: the seven-million-parameter cone codec of E-069, which learned persistence from history and did not learn velocity.

Every rung after it is bought only on the previous rung's read-out, under the frozen protocol — train on data up to 2020, test on 2021 to 2024, with longitude holes — and with the NULL LADDER beside it. A null ladder is the set of deliberately dumb forecasts a real one has to beat.

Those are: climatology (the long-term average for this day of the year), persistence (tomorrow equals today), damped persistence (today's anomaly shrunk towards zero), the LIM (the plain linear forecast fitted to the same field, which today beats every learned head from fifteen days out), nearest-analogue retrieval, and the rung below.

Vocabulary for the recipe. PERCEIVER is an architecture that cross-attends a variable, unordered set of inputs into a small fixed set of latent vectors, so cost does not grow with the number of inputs — the right shape when the input is a bag of dots.

A MIXTURE OF EXPERTS replaces one feed-forward block with many and lets a router send each token to two of them, so the model can hold billions of parameters while paying only for the fraction that fires; here the router is told which sphere the token sits on, which makes the specialisation natural.

A GAUSSIAN NEGATIVE LOG-LIKELIHOOD asks the model for a mean AND a spread and scores both, so a confident wrong answer is punished harder than an honestly uncertain one. WARM-UP ramps the learning rate up from zero over the first two thousand steps so early gradients cannot wreck the weights; COSINE then eases it down to a tenth of the peak.

μP is a parametrisation that scales the per-layer learning rates with width, so the best rate found on a small model stays best on a large one — that is what makes a multi-billion run one shot instead of a search. Z-NOISE is deliberately adding noise to the embeddings the forecaster reads, which turns forecasting into denoising.

MEASURED in this programme: the +0.045 and +0.050 that z-noise buys, the 3e-4 and 4e-4 learning rates, the flat width ladder, and the failure of H1 — velocity from displacement — at five seeds. ASSUMED: every hour count and token count in the table, the rung sizes, the four-epoch limit, and the mixture-of-experts design itself. Nothing above L1 is dispatched.

Sources [E-072 plan — the Earth foundation model: data design first, then the scaling ladder, then the recipe](#) · [Handover, 4 Sep 2026 — cone v2, the four spheres, the Earth foundation model, and where to fork](#) · [ml/SCALING.md — how skill moves with model and data size \(the programme's scaling audit\)](#) · [Hoffmann et al. 2022 — Chinchilla: training compute-optimal large language models \(~20 tokens per parameter\)](#) · [Muennighoff et al. 2023 — scaling data-constrained language models \(the \$\leq 4\$ -epoch limit\)](#) · [Yang et al. 2022 — Tensor Programs V \(\$\mu P/\mu Transfer\$: tuning the learning rate on a small model\)](#) · [Perceiver IO](#) · [E-069 plan — two stencils, one cone](#) · [E-069 in the experiment log](#) · [E-070 plan — the global tensor \(family 7\): the globe at 0.25° pentads, gates G1–G3, phases A–F](#) · [E-071 plan — cone v2: global, harmonic, Argo-aware, sized by the fastest thing in the ocean](#) · [Earth 2 paper \(v8, PDF\)](#) · [ml/CLAUDE.md \(replication rules, §3b\)](#)

Read-outs: El Niño, AMOC, colour — and what each needs first

A READ-OUT is something we want to know: the strength of the overturning, the state of El Niño, the colour of the sea. A HEAD is the small model that reads it out — a probe of a few hundred thousand parameters trained on top of a frozen embedding.

The band at the top of the slide is the point of the whole programme: because every read-out is a head on ONE representation plus the rolled forecast, adding a read-out costs a probe and not a training run. The read-outs are the cheap, replaceable end of the system; the forward model in embedding space is the programme.

Row by row. EL NIÑO is scored on Niño-3.4 — the sea-surface temperature anomaly averaged over a box in the central-eastern tropical Pacific — and on the ONI, its three-month running mean, the official index the US Climate Prediction Center publishes. Our own bake of the OISST record puts Niño-3.4 at -0.57 in January and $+2.09$ in July 2026, and that has to be reconciled with CPC's number before it can serve as a label.

Among its nulls is the SPRING BARRIER: forecasts made across the northern spring lose skill much faster than forecasts made at other times of year, because the coupled ocean-atmosphere signal in the tropical Pacific is weakest then. So an honest ENSO forecast is a skill curve by start month, not one number.

And without the subsurface — the warm-water volume, or the depth of the $20\text{ }^{\circ}\text{C}$ isotherm — a head on surface temperature alone is a persistence forecast wearing a name. AMOC is scored against the RAPID array at 26.5° north, then MOVE, OSNAP, SAMBA and the Florida cable; those truth series are already in our tensor.

What AMOC needs is the buoyancy forcing — the heat and fresh water the atmosphere puts into the ocean — and the three-dimensional interior. SURFACE COLOUR needs the reflectance bands as inputs with chlorophyll as the target, never as an input: that is the observed-over-derived rule from earlier in this deck.

Sea ice, land water and the atmosphere are the cheap three: they need only Phase L0, six gap-free channels from two sources, about forty gigabytes — which is why they sit at the top of the to-do list on the right even though they are not the headline.

MEASURED here: the Niño-3.4 values from our own OISST bake, and the fact that the RAPID and Florida-cable series are already in the tensor. ASSUMED: every prerequisite ranking and every null. No head on this slide has been trained.

Sources [E-072 plan — the Earth foundation model: data design first, then the scaling ladder, then the recipe](#) · [Handover, 4 Sep 2026 — cone v2, the four spheres, the Earth foundation model, and where to fork](#) · [E-071 plan — cone v2: global, harmonic, Argo-aware, sized by the fastest thing in the ocean](#) · [E-070 plan — the global tensor \(family 7\): the globe at \$0.25^{\circ}\$ pentads, gates G1-G3, phases A-F](#) · [ml/plans/DATA_LADDER.md \(source\)](#) · [CONE DATA AND ENSO.md — the companion page](#) · [IRI/CPC ENSO forecast plume](#) · [Oceanic Niño Index \(NOAA CPC\)](#) · [PMEL Warm Water Volume / \$20\text{ }^{\circ}\text{C}\$ isotherm](#) · [GODAS ocean analysis \(NOAA PSL\)](#) · [RAPID-MOCHA-WBTS \$26.5^{\circ}\$ N data](#) · [Florida Current cable transport \(NOAA AOML\)](#) · [ESA OC-CCI](#) · [PACE](#) · [OSI SAF sea-ice products](#) · [GRACE/GRACE-FO JPL mascons](#) · [ESA CCI soil moisture](#) · [ERA5 \(CDS\)](#) · [GPM IMERG](#) · [ECMWF open data](#) · [E-062 in the experiment log — the flat width ladder and the Argo information finding](#) · [E-066 in the experiment log — the 200-mode linear inverse model as the reference forecast](#) · [ml/CLAUDE.md \(replication rules, §3b\)](#)

Phase 0 first — the decisive experiment fits the data we already have

The plan starts with an experiment that needs no new data.

Phase 0 uses exactly the tokens the second stage already reads: ocean pixel-months, the 3-by-3 codec, and the sunflower stencil over 24 months. We train a cone codec over those same tokens with the masked-dot objective, and compare three things at equal total context: the current codec with its stage-2 attention head; the cone codec with a plain ridge regression on top; and the cone codec with an attention head. Three things about the protocol are non-negotiable after the 2 September paper reset. The verdict is rolled skill under the corrected window-scope pool — mean skill against climatology and against damped persistence at each lead, trained and held-out longitudes reported separately, with block-bootstrap intervals and the Linear-Inverse-Model null beside every number. Probes — the velocity probe and the RAPID transport probe — are diagnostics of what the embedding encodes, never verdicts. And at least three seeds per arm, because anything probe-scored is unreadable at one seed. The predictions are written down so they can be wrong: the cone codec should match or beat the current setup on rolled skill with only a linear head, and beat the embedding-space LIM where the current codec does not; it should win by a wide margin on the velocity probe; and it should tie on static probes. If the cone codec only matches when it also gets an attention head, then the "work in the embedding" argument loses on sample efficiency and the hybrid keeps a thick stage 2.

After that, one sphere per phase, in the order agreed earlier. Phase 1 adds the ocean surface and the atmosphere as forcing — the gap-filled surface maps and ERA5 — and is scored on forecasting sea temperature and sea level one to four weeks ahead against persistence and against the operational Copernicus forecast. Phase 2 adds the interior — Argo as native dots, GLORYS as a flagged, heavily masked prior — and is scored on the AMOC again, on mixed-layer depth, and on upper-ocean heat content. Phase 3 adds the ocean biosphere and is scored on reconstructing the carbon flux with the carbon maps held out. Phase 4 adds land, with the local codec on Sentinel and Landsat tiles, and is scored on vegetation and soil-moisture forecasts as well as the standard benchmarks for comparability. Phase 5 adds the fast atmosphere at ten-minute cadence through a second local codec.

Six ablations run in every phase; the first — cone inside versus cone on top at equal context — is the one that keeps answering the zero-sum question as the inputs grow. The engineering note at the bottom is a warning: the network is cheap, under an hour per year of ocean embeddings on a single modern accelerator; the sampler that finds which tiles, dates and profiles fall into which cones is the thing to build first.

Sources [Generic-embedding input proposal \(full note\)](#) · [Earth 2 paper \(v8, PDF\)](#) · [ml/CRAUDE.md \(replication rules, §3b\)](#) · [Copernicus global analysis/forecast](#)

Data continuity 2026-27 — build the archive sensor-agnostic

The verification pass this week turned up an unusual amount of change in the satellite fleet, and it affects which channels are safe to build on.

Sentinel-1 lost its A satellite at the end of June; the constellation is now C and D, and there is a two-year single-satellite gap in the radar record. Sentinel-2 has three satellites only until the end of this year. MODIS, the sensor behind most 500-metre products since the year 2000, is being switched off around the turn of the year, and Suomi-NPP, the first VIIRS satellite, stops delivering data on the first of November; the newer NOAA-20 and NOAA-21 continue. The passive-microwave sea-ice record is in the middle of a hand-over from the retiring military DMSP satellites to Japan's AMSR2 and, eventually, AMSR3 — which means that for a while near-real-time sea ice depends on a single satellite launched in 2012. Sea level is stable; Sentinel-6B is up but not yet the reference. Several climate-quality records have hard ends — the ocean-colour climate record at the end of 2024, the FLUXNET collection frozen in 2020 — and their near-real-time twins must be used for later years.

The architectural answer is the one the whole deck has been building towards: with channel-drop training, a sensor that disappears is a masked channel, not a failure. But the training archive has to be assembled sensor-agnostic, with the sensor identity in the source flag, so that the model learns "thermal radiometer at 1 km" rather than "MODIS".

The licence line matters for anything meant to be published: the EN4 ocean analysis is non-commercial, the river-gauge archive is research-only and cannot be redistributed, the European radar composite is for members, and ECMWF's native 9-kilometre analysis is licensed while the quarter-degree open data is free.

The paragraph at the bottom lists smaller corrections that came out of the same pass — worth knowing because they are commonly assumed the other way: ERA5's sea-surface temperature is an input, not a product; ERA5-Land has no data assimilation; the Copernicus wave model is MFWAM, not WAVEWATCH; and the European radar composite is now 1 kilometre every 5 minutes.

Sources [Generic-embedding input proposal \(full note\)](#) • [Sentinel-1 orbital reconfiguration \(S1A end\)](#) • [Sentinel-1D data opening](#) • [Sentinel-2A extension to end-2026](#) • [NASA MODIS → VIIRS transition](#) • [NOAA: Suomi-NPP data end 1 Nov 2026](#) • [NSIDC: SSMIS sunsets, AMSR2 rises](#) • [NOAA: AMSR2 → AMSR3](#) • [Sentinel-6](#) • [ESA OC-CCI](#) • [ESA CCI soil moisture](#) • [EN4 \(Met Office\)](#) • [GRDC river discharge](#) • [EUMETNET OPERA](#) • [GLEAM](#) • [ECMWF open data](#)

El Niño 2026 — what the catalog predicts, and what to add

El Niño is the warm phase of a see-saw in the tropical Pacific called the El Niño–Southern Oscillation. Normally the trade winds blow from east to west, piling warm water up near Indonesia and letting cold water rise off South America. When the trades weaken, that pile sloshes back east, the eastern Pacific warms, and the world's rainfall patterns move with it. NIÑO-3.4 is the agreed thermometer: the average sea-surface temperature anomaly in the box from 5° south to 5° north, 170° west to 120° west. The ONI, or Oceanic Niño Index, is the official version of that number — a three-month running mean computed on a different, coarser temperature reconstruction — and an event is declared when it stays above +0.5 °C for five overlapping seasons.

The chart on the left is computed here, from the repository's own baked temperature archive against its 1991–2020 normal, and it shows the fastest possible development: mildly cool in January, crossing zero in March, and +2.09 °C by July. The three dashed lines are the December peaks of the three biggest recent events from the same bake. Our number for July is 0.7 °C above the Climate Prediction Center's, entirely because of the different dataset and the running mean — which is the argument for carrying the official index alongside our own rather than instead of it.

Why sea-surface temperature alone forecasts poorly beyond about three months: the memory of this system is not at the surface, it is in the THERMOCLINE, the sharp boundary a hundred-odd metres down between warm surface water and the cold deep. WARM WATER VOLUME — how much water above 20 °C is stacked up across the equatorial Pacific — leads the surface signal by roughly six months, and it is the single most valuable series absent from our catalog. On top of that slow memory sit fast triggers: a WESTERLY WIND BURST, a one-to-two-week reversal of the trade winds usually set off by the Madden–Julian Oscillation, can launch an eastward wave that tips a primed ocean into an event. And there is the SPRING PREDICTABILITY BARRIER: forecasts made in boreal spring are markedly worse, because the signal is weakest and the noise strongest exactly then. That is the barrier any honest skill curve has to be plotted across, and the reason the ranked additions on the right put the subsurface first and the daily winds second.

Sources [CPC ENSO diagnostic discussion, 13 Aug 2026](#) · [IRI/CPC ENSO forecast plume](#) · [PMEL Warm Water Volume / 20 °C isotherm](#) · [GODAS ocean analysis \(NOAA PSL\)](#) · [NOAA interpolated outgoing longwave radiation](#) · [Oceanic Niño Index \(NOAA CPC\)](#) · [NOAA PSL climate indices](#) · [Multivariate ENSO Index v2 \(NOAA PSL\)](#) · [RMM Madden–Julian Oscillation index \(BoM\)](#) · [CCMP ocean vector winds \(REMSS\)](#) · [OSCAR L4 final v2.0 \(PO.DAAC\)](#) · [data/catalog.json — the 274-record catalog \(source\)](#) · [docs/CATALOG.md — the catalog in prose](#) · [CONE DATA AND ENSO.md — the companion page](#)

Family 8: the measurement is never missing, only far away

Some vocabulary first. The CONE is the shape of everything the forecaster is allowed to read when it predicts one cell at one moment: nearby places at recent times, and places further away only at times further back, because a signal needs time to travel. The SUNFLOWER is how we place the sample points inside that cone — a spiral of dots around the anchor, dense near it and sparse far from it. A TOKEN is one item in the list the model reads; the anchor's whole input is a few hundred of them. A MISS TOKEN is what we hand the model when a channel has no value at the place and time it asked about: a marker that says "nothing here", carrying no number.

An ARGO FLOAT is a drifting robot that sinks to two kilometres, drifts for about ten days, then rises taking temperature and salinity all the way up and radios the profile home. There are roughly four thousand of them. They are the only routine measurement of the ocean's interior — and they are points, not a map. To fit them into a grid today we take somebody else's smoothed monthly one-degree reconstruction of those points, which means the model never sees a measurement, only a map of one, and at most bins the map is simply absent.

Family 8 proposes the opposite. Instead of asking "what is in this grid cell", ask "which are the k nearest actual observations, and how far away are they" — the K -NEAREST-NEIGHBOUR idea. Each token then carries the measured value plus its offset east, its offset north, how many days ago it was taken, and how crowded that neighbourhood was. Distance becomes information rather than an excuse: a reading eighty kilometres and three days away deserves more weight than one four hundred kilometres and three weeks away, and the model can learn that by itself.

Three guards keep it honest. ONE-SIDED IN TIME means the search only ever looks backwards from the anchor's own five-day bin, so a forecast can never quietly read its own future. BOUNDED means each channel declares a maximum radius and a maximum age, so the search cannot wander across an ocean to find something; only when that bounded search truly comes up empty does the old miss token appear, and then it is a measured fact. And k is set by a CORRELATION LENGTH — the distance over which a field stays recognisably similar to itself, about three hundred kilometres near the surface — rather than by taste. The Argo arithmetic in words: four thousand floats each surfacing every ten days is about two thousand profiles in any five-day window over the whole ocean; spread over three hundred and sixty million square kilometres that is one profile per hundred and eighty thousand square kilometres, so the nearest one is typically a couple of hundred kilometres away. Allow thirty days instead of five and you have six times as many, the nearest falls to about ninety kilometres, and the fifth nearest sits at about two hundred — one correlation length. Hence $k = 5$ within thirty days.

What it is NOT is a replacement for the cone. Dense channels — currents, sea-surface temperature, the reanalysis — are already complete maps, and the sunflower's whole point is that it samples them at several scales at once; five neighbours would collapse that to the three cells nearest the anchor. So family 8 is a HYBRID: an array for the dense groups, plus an OBSERVATION STORE — a plain list of (time, latitude, longitude, channel, value) rows — and a precomputed neighbour index beside it. That is also why it stops being a single tensor, and it is the door to feeding raw profiles, moorings and satellite tracks in without gridding them first.

Sources [E-071 plan — cone v2: global, harmonic, Argo-aware, sized by the fastest thing in the ocean](#) · [E-070 plan — the global tensor \(family 7\): the globe at 0.25° pentads, gates G1-G3, phases A-F](#) · [Argo](#) · [Roemmich-Gilson Argo climatology \(Scripps\)](#) · [Aardvark Weather \(Nature 2025\)](#) · [GraphDOP](#) · [ESFM](#) · [CONE DATA AND ENSO.md — the companion page](#)

Family 7 vs family 8, side by side — and the one experiment that decides

This slide puts the two designs in one table so the difference is a row-by-row comparison rather than an argument.

Read the middle column as the world we have. A sparse channel arrives as somebody's gridded map; at an anchor we take the nearest one-degree cell; that cell holds a number in eight per cent of the five-day bins and nothing in the rest, so the model mostly reads a miss token. In the right-hand column the same channel always produces exactly k tokens, each one a real measurement with its distance and its age attached. Everything about the dense channels is identical in both columns — that is the point of calling family 8 a hybrid rather than a replacement.

Two rows deserve a gloss. A KD-TREE is a simple data structure that sorts points by position so that "which five are nearest to here" is answered in microseconds instead of by scanning every point; building one per five-day bin makes the whole neighbour lookup a one-off cost. And the ANOMALY row matters more than it looks: an anomaly is the departure from what is normal for that place at that time of year, and today we compute it at the grid cell, whereas family 8 would evaluate the same seasonal fit at the observation's own latitude, longitude and day — closer to the truth, and free.

The box at the bottom is the whole proposal reduced to one cheap test. A PREDICT-THE-MEAN BAR is the dumbest possible forecast: ignore the input and always answer with the long-run average. Any model has to beat it, and the reason we score per family of channels against that bar is that a headline loss can look fine while one group of channels is quietly contributing nothing — measured against the bar, a group that has learned something separates from a group that has not. The falsifier is stated in advance: if replacing the thirty-two gridded Argo channels with five profile tokens does not lower the loss on hidden interior channels compared with an otherwise identical run, then distance-as-a-feature bought nothing and family 8 does not get built. That costs about one seed, under two hours on a single consumer graphics card.

The last sentence is the one not to lose. An earlier experiment removed the Argo targets altogether — stopped asking the model to predict the ocean interior — and the forecast got WORSE, not better. A channel whose removal hurts is a channel carrying signal, which means the interior is worth representing properly; the question family 8 asks is only whether we are currently representing it in the stupidest available way.

Sources [E-070 plan — the global tensor \(family 7\): the globe at 0.25° pentads, gates G1-G3, phases A-F](#) · [E-071 plan — cone v2: global, harmonic, Argo-aware, sized by the fastest thing in the ocean](#) · [E-069 in the experiment log](#) · [ml/plans/DATA_LADDER.md \(source\)](#) · [Argo](#) · [Roemmich-Gilson Argo climatology \(Scripps\)](#) · [ml/CLAUDE.md \(replication rules, §3b\)](#) · [Earth 2 paper \(v8, PDF\)](#)

Sources

Every paper, model card and repository the deck draws on, grouped by model and topic, as short URLs. A few notes on how they were used.

For the six model slides, numbers were taken from the papers and, where possible, verified against the published configuration files: Prithvi-EO 2.0's token sizes come from its Hugging Face config.json (16-pixel patches for the 300 M model, 14 for the 600 M), and Granite-Geospatial-Ocean's architecture from the config and model file in the ibm-granite GitHub repository (42-pixel tiles, 17 channels, embedding 512, 12 layers, patch size 2 by 2, one frame).

The head-to-head tables were transcribed from the papers' own tables: OlmoEarth Tables 2, 3 and 7; TerraMind Table 6; TESSERA Table 1; Granite Table 1; OceanSAR-1 Tables 1–3; GEO-Bench-2 for independent ranks; and an independent Local Climate Zone study for AlphaEarth versus TESSERA.

For the ocean-including models and related work, sources were checked in August 2026 and include several 2026 papers (AIFS surface ocean, ESFM, SamudrACE, PARADIS, ACE2-NEMO). Earth 2 numbers — the attribution matrix and probe ladder — are quoted from the project's own paper draft.

Items listed as "also checked" are things we looked at and excluded, with the reason, so that nobody re-chases them.

Sources [Survey notes](#) • [Generic-embedding input proposal \(full note\)](#) • [Earth 2 paper \(v8, PDF\)](#)